

Minor Project



Project Title: Optimal selection and design of off-Grid Hybrid renewable energy system

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List of Contents



Sr. No	List of Contents
1.	Introduction
2.	Literature Review
3.	Problem Statement
4.	Objectives
5.	Methodology
6.	Proposed System
7.	Results and Conclusions
8.	References

Introduction

- Why renewable energy sources?
- Shifting focus to other renewable energy sources.
- Simulation and optimization techniques, grid-connected and off-grid renewable (wind and solar)/hybrid energy systems are developed and implemented for electricity generation
- Basically, now the future is integrated Renewable Energy system not PV alone

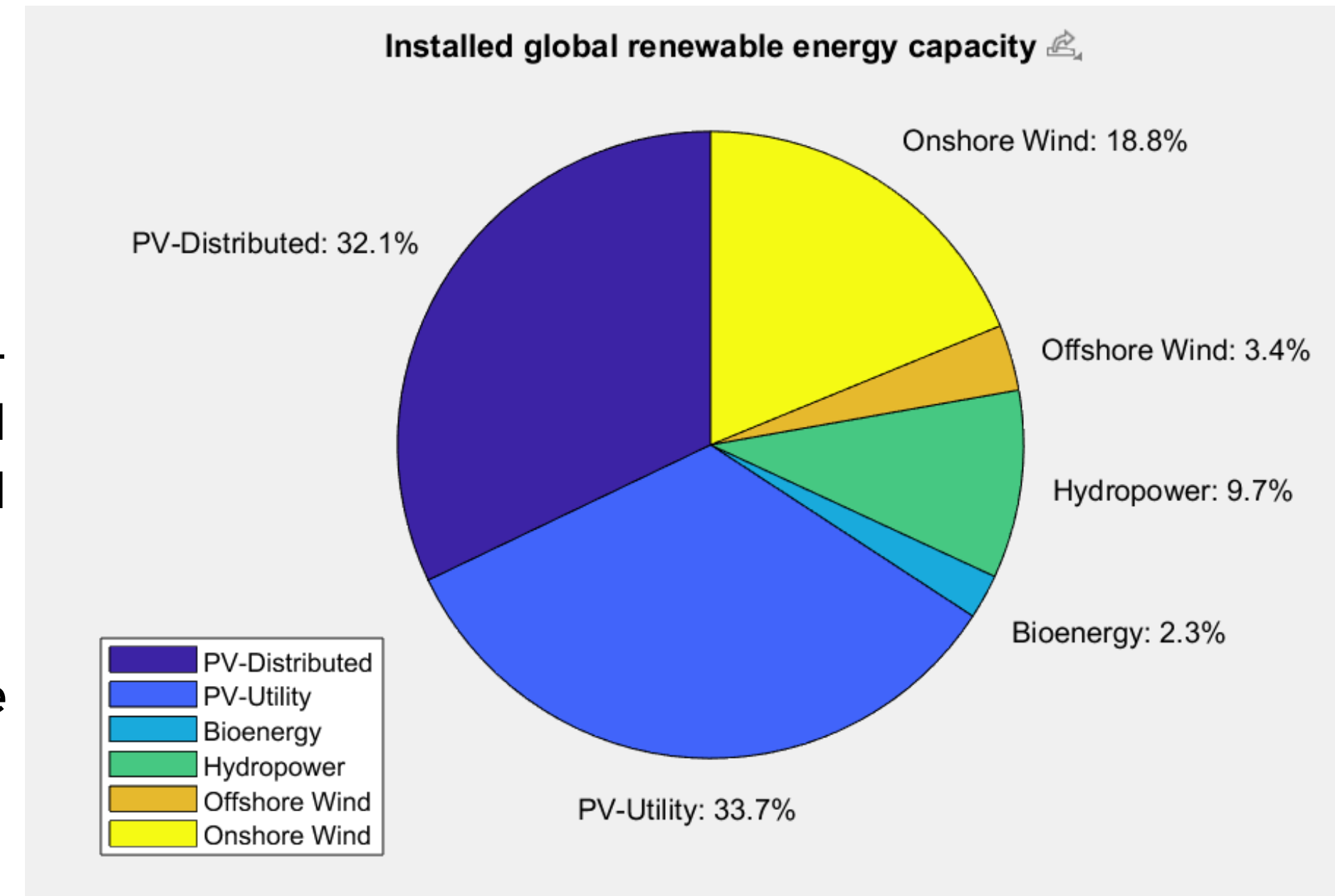


Fig 1 - Renewable energy statistics 2024 ,The International Renewable Energy Agency (IRENA) [1]

Total Global Renewable Energy Distribution; PV and Wind Energy sources leading the contribution

Introduction

- India plans to use 500 GW of installed capacity, mostly from PV and wind energy sources, to reach 50% of its energy coming from renewable sources by 2030 at COP 26 [1]
- India is a leading player in the race to implement RESs for the Electricity demands
- Moreover, the focus is now getting shifted more towards the integrated systems of different renewable energy
- Various combinations and models are being researched upon and their Tecno-Economic viabilities being judged namely;
 - i) PV-Standalone, PV-Grid Connected
 - ii) PV+ Wind Turbine + Battery System
 - iii) PV+ Wind Turbine + Diesel Generator
 - iv) PV+ Wind Turbine+Battery+Diesel Generator

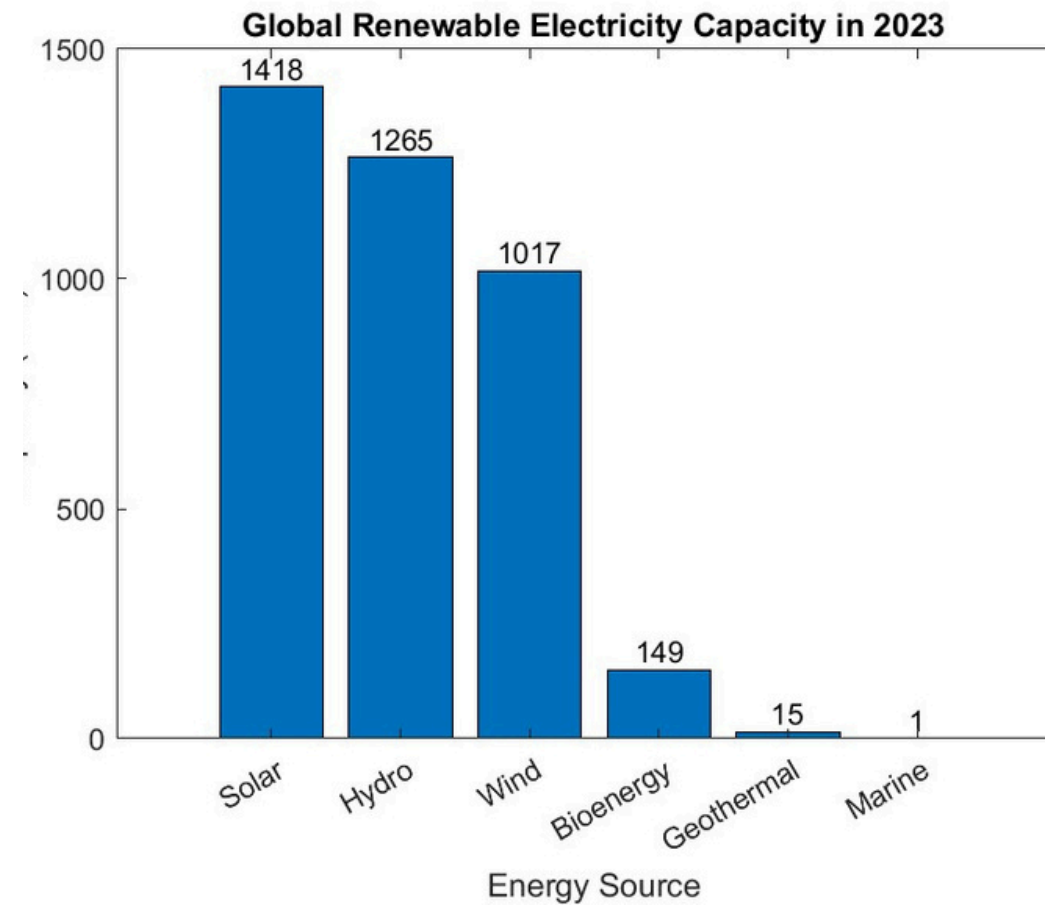


Fig 2 - Renewable energy statistics 2024 ,The International Renewable Energy Agency (IRENA) [5]

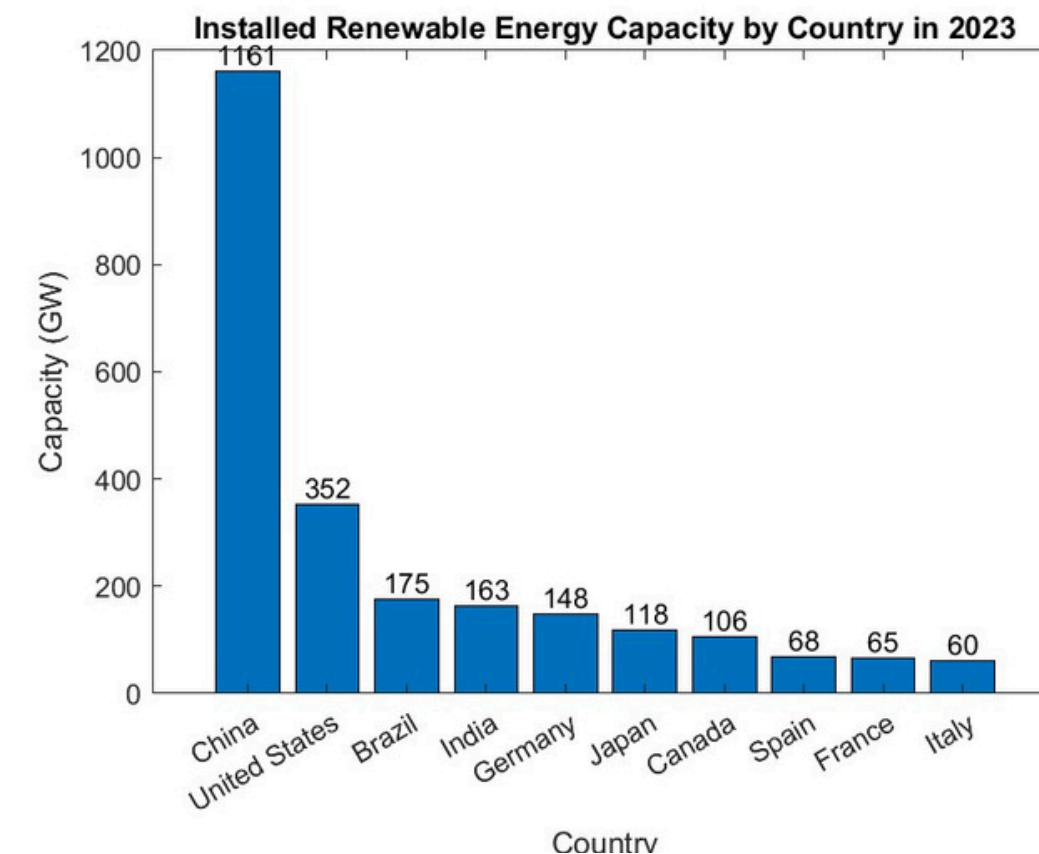


Fig 3 - Renewable energy statistics 2024 ,The International Renewable Energy Agency (IRENA) [5]

Literature Review			
Sr. NO.	Journal and Year of Publication	Authors	Title
1.	ScienceDirect International Journal of Hydrogen Energy <u>Volume 21</u> , March 2024, 101964	Mansoor Urf Manoo, Faheemullah Shaikh, Laveet Kumar, Müslüm Arıcı	Comparative techno-economic analyses and optimization of standalone and grid-tied renewable energy systems for South Asia and Sub-Saharan Africa.
2.	Elsevier , Sustainable Cities and Society May 2024, 105291	Muhammad Ishaq, Ibrahim Dincer	Investigation of a tri-renewable energy system coupled with battery and hydrogen storages for a sustainable city.
3	Elsevier , Scientific African <u>Volume 25</u> , September 2024, e02305	Chidozie Ezekwem, Suresh Muthusamy, Peace Chiamaka Ezekwem ..	Optimal selection and design of grid-connected hybrid renewable energy system in three selected communities of Rivers State.
4.	Elsevier , International Journal of Hydrogen Energy, Volume 46, Issue 70 11 October 2021	Srijit Basu, Alfred John, Akshay, Anil Kumar.	Design and feasibility analysis of hydrogen based hybrid energy system: A case study.

Literature Review

Result and conclusions of paper 1.

- This research explores the techno-economic feasibility of hybrid power plants to provide electricity in rural and underserved areas of India, Myanmar, Ghana, and Cameroon. It identifies optimal hybrid systems, including PV/WT/DG/BAT for India, Ghana, and Cameroon, and WT/DG/BAT for Myanmar. India's Levelized Cost of Energy (LCOE) is \$0.127/kWh for stand-alone systems and \$0.026/kWh for grid-tied systems, with significant cost savings reflected in the Net Present Cost (NPC). The study highlights the crucial role of sound energy policies in improving energy access, reducing costs, and promoting renewables in South Asia and Sub-Saharan Africa.[1]
- Khiriya Bharka, a small village in Isagarh tehsil, Ashoknagar region, Madhya Pradesh, has 320 residents and 80 houses. The electrical load includes domestic appliances like lights, fans, mobile chargers, and televisions. Public buildings, such as schools, community halls, health clinics, and streetlights, contribute to the load. The residential and community demand for the 80 houses is estimated to be 286.95 kWh/day, with a peak load of 28.67 kW, designed to handle summer's higher peak demand. Commercial loads from stores and flour mills, consisting of lights, fans, and motors, require 262.15 kWh/day, with a peak load of 87.53 kW.

System Combination	PV (kW)	WT (EO20)	Gen (kW)	1 kWh LI	Converter (kW)	NPC (\$) Cost	LCOE (\$/kWh) Cost	Operating cost (\$/yr.)	Total Fuel (L/yr.)
PV/WT/DG/BAT/ Converter	63.3	2	130	235	95.9	\$328,791	\$0.127	\$8159	1455

Table 1 - Result summary of the optimal combination for selected place Khiriya Bharka, India. [1]

Literature Review

Result and conclusions of paper 2.

- This research explores the Potential solutions in terms of energy storage options are imperative to create a balance between the locally available RES and the load profile of the community because energy storage is a key factor in the continuity of energy supply to the community. Among different energy storage options, hydrogen storage is considered to be a crucial option because of its high energy density, and hydrogen being the most common element in the universe is incarnated with satisfactory energy storage capability.[2]
- The community of Inuvik contains around 3214 people in 2022. The community’s average electrical power requirement is around 3.47MW and the annual energy needs are around 30600 MWh.

Parameters	Values	Unit
Inuvik population for 2022	3214	Persons
Hot water requirement per person	75	Liters
Annual hot water requirement	87983.2	Kiloliters
Annual heat requirement	97200	MWh
Annual energy needs	30600	MWh
Total number of average persons in a household	2.7	People
Average number of vehicles per household	1.5	Vehicles
Total number of households	1190	Households
Total number of vehicles in Inuvik	1786	Vehicles

- Load directly covered by solar PV 7435.8 MWh
- Load directly covered by wind energy 3182.4 MWh
- Load directly covered by biomass
- 19981.8 MWh

Table 2- Potential needs of the community residing in the Inuvik region of Canada. [2]

Literature Review

Result and conclusions of paper 3.

- This study evaluates electrification and performs a techno-economic analysis of hybrid systems for rural communities in Rivers State. The optimal choice, based on hourly electricity demand, is a grid/PV module combination. The proposed systems have a lower cost of energy (COE) compared to the grid, and the research includes a sensitivity analysis to guide decision-making and system design.[3]
- The Aluu, Choba and Rumuekini communities experienced an average daily electricity supply of 4.6, 5.4 and 6.2 h respectively from the national grid with energy usage of 5436.4 MW for Aluu, 3375.4 MW for Choba and 1626.1 MW for Rumuekini, in 2022.
- The cost of energy (COE) for the proposed systems ranged from 0.0181 \$/kWh to 0.0185 \$/kWh, significantly lower than the grid’s COE of 0.08625 \$/kWh in the respective communities.

Production	kWh/yr	%
SunPower	42,822,730	70.9
Grid Purchases	17,554,098	29.1
Total Consumption	60,376,828	100
AC primary load	30,627,150	55.5
Grid sales	24,575,656	44.5
Total Quantity	55,202,806	100
Excess electricity	3192,511	5.29

Table 3.a-Annual electrical production and utilization of the optimized C1 system in Aluu[3]

Types	values	NPC (\$)	LCOE (\$/kWh)
PV cost (\$)	652.4	32.5	0.0142
Discount rate (%)	9.775	41.2	0.0141
Inflation rate (%)	13.345	42.8	0.0249
Scaled solar irradiance (kWh/m2_)	3.5105	51.4	0.0231
Electrical demand (kWh/day)	79,615	40.1	0.0185

Fig 3.b-Annual electrical production and utilization of the optimized C1 system in Aluu [3]

Literature Review

Result and conclusions of paper 4.

- The study focuses on the issues of hydrogen energy storage which can solve the fluctuating output power problem by simulating results on HOMER software. Three combinations of the Solar-Hydrogen system, Wind-Hydrogen system, and Solar-Wind-Hydrogen hybrid system are presented to find the most optimum one. Levelized Cost of Energy (LCOE) for Hybrid System has proven to be the most economical while the Wind Turbine cost 1.476% higher and the Solar Photovoltaics (PV) System costs 108.03% more. LCOE for Hybrid Model is \$0.3387, while for Solar System it is \$ 0.7046 and for Wind System it is \$ 0.3437. These results show that a hydrogen-based energy storage system is viable for the considered.
- Main purpose of the system is the electricity supply of a residential community and hydrogen production to stabilize the intermittency of renewable energy sources. Simulations were done using HOMER software for power stations energized by Solar-Hydrogen storage, Wind-Hydrogen storage, and Wind and Solar Hybrid-Hydrogen storage power system, The software HOMER is used to detect the most feasible electrical load component sizes.

Economic parameter	Solar PV system	Wind turbine system	Hybrid system
Net present cost (\$)	3,379,547.00	1,663,756.00	1,638,635.00
CAPEX (\$)	605,683.59	313,220.66	308,095.88
OPEX (\$)	78,610.08	35,034.72	34,759.88
LCOE (per kWh, \$)	0.7046	0.3437	0.3387

Table 4- Economic summary from all the systems [4]

Problem Statement

Note:

- Throughout this project we'll be focusing on an Indian Village Khiriya Bharka. Khiriya Bharka is a small village in the Isagarh tehsil, Ashoknagar region, Madhya Pradesh.[Fig-4]
- Madhya Pradesh is enriched with natural energy resources, including solar and wind energy
- In Khiriya Bharka, there are 320 residents overall, including 80 houses. On the domestic load, there are lights, fans, mobile chargers, and televisions.
- The residential and community load demand for 80 identical houses is estimated to be 2.86.95 kWh/day, with a peak load demand of 28.67 kW
- Fig 5 shows the daily load profile of Khiriya Bharkha

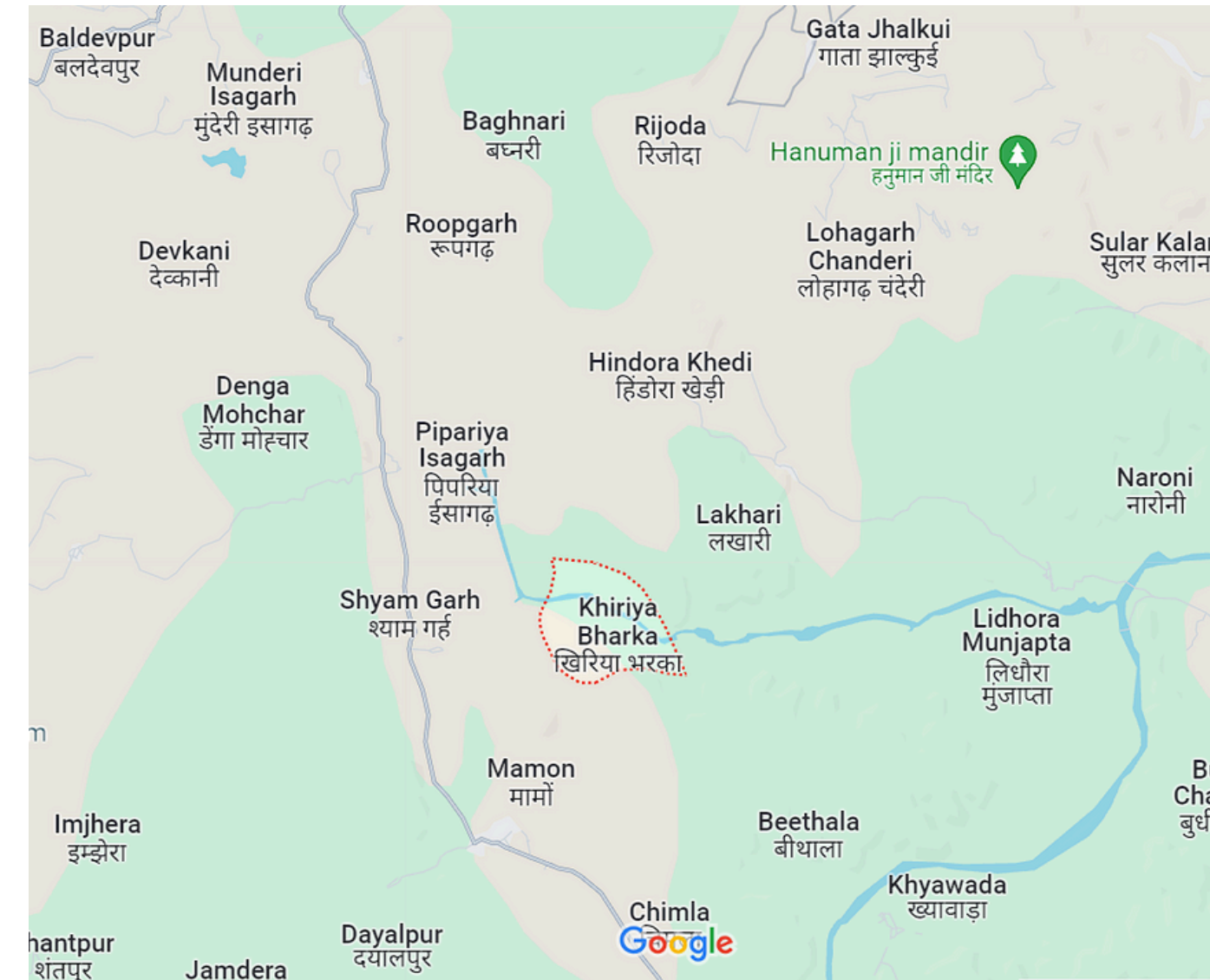


Fig 4 - Khiriya Bharka, Madhya Pradesh 473446, Google Maps[7]

Problem Statement

- Energy Resources are also abundant in this village
- Hence making it feasible for us to design a solar and wind integrated sytem
- The estimated yearly average solar radiation data for Khiriya Bharka, India, is 4.82 kWh/ m²/ day and the average clearness index is 0.536 per month
- In Khiriya Bharka, the maximum windy month is May, and the yearly average wind speed is 10.41 m/s determined at a maximum 50-m height range.
- Fig 6,7 shows the Solar and Wind specifications of the village respectively

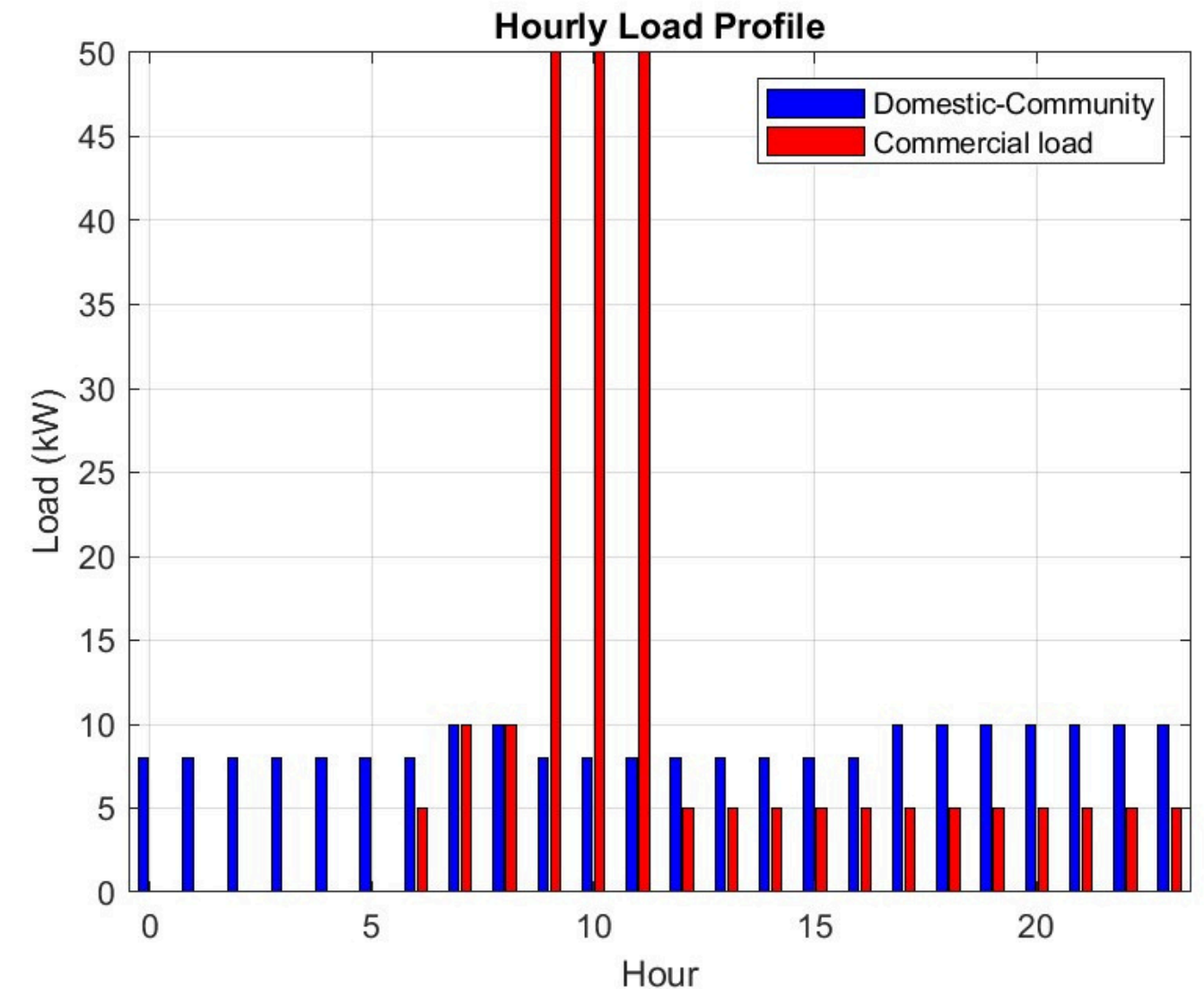


Fig 5- Daily Load Profile for both Domestic and Commercial

Problem Statement

Issues associated with the existing HRES

- PV Systems come with their own problems and same is the case with Wind turbine systems
- Even the integration of PV and WT systems are not capable of meeting the total demands of any locality
- In order to design a complete standalone system we are compelled to add conventional Diesel Generators in order to meet the load demands
- Another configuration that is commonly studied during literature review was PV+WT+BAT, which again has several shortcomings of its own
- A comparative study of all these Combinations is summarized in Figures shown in pages ahead, With results like NPC and LCOE

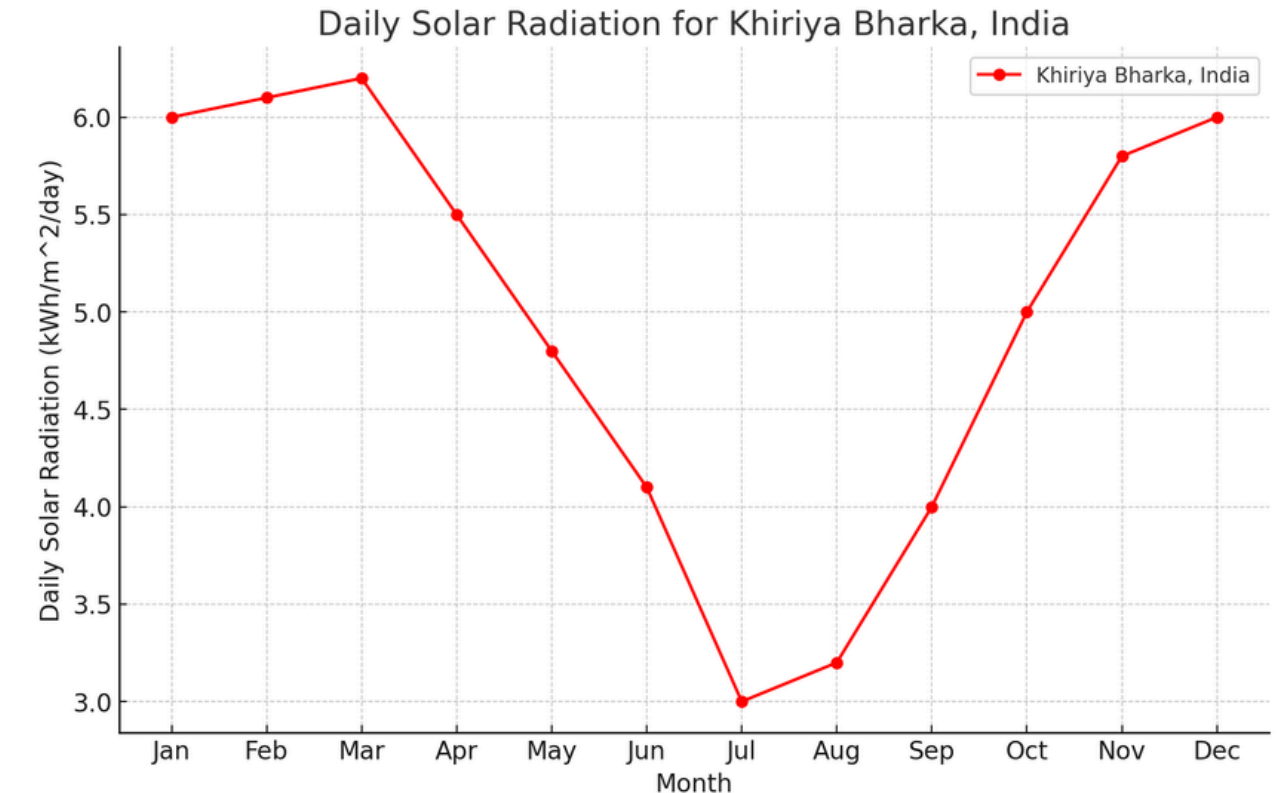


Fig 6 - Monthly average of Solar Radiations over the year[6]

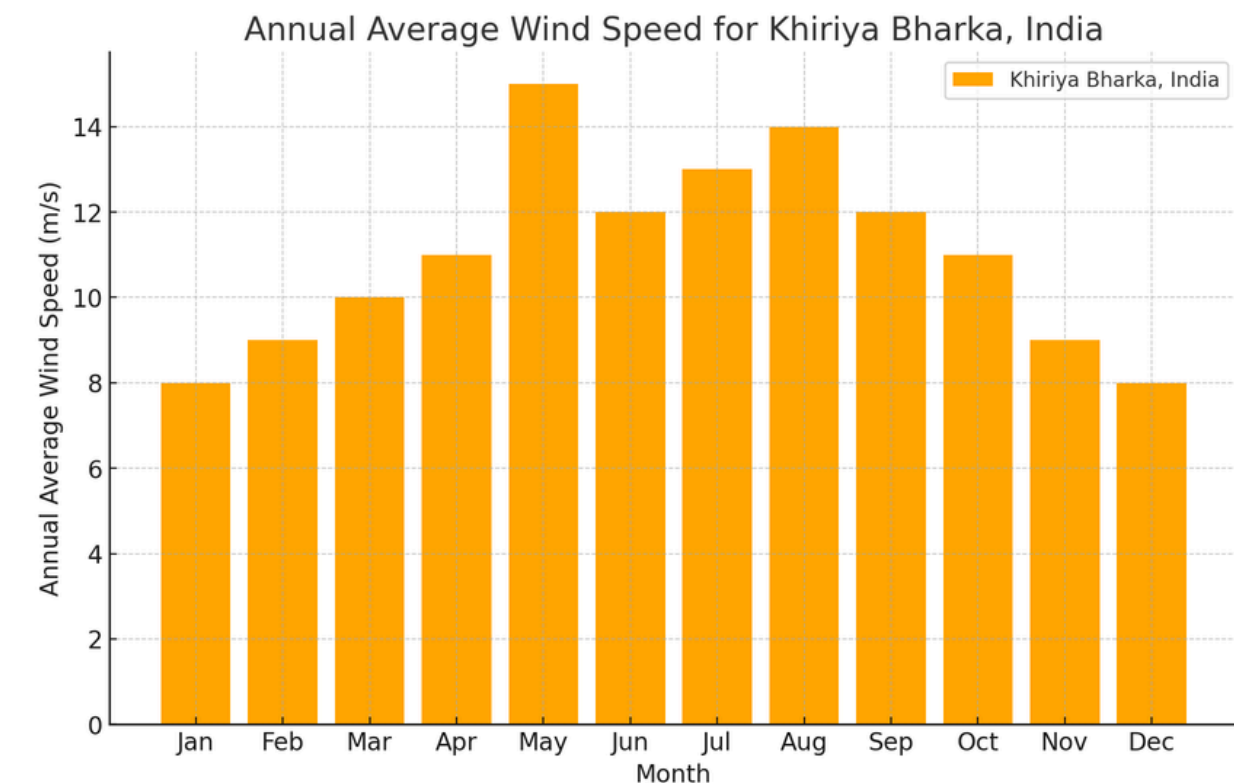


Fig 7 - Annual average wind speed over the year[6]

Problem Statement

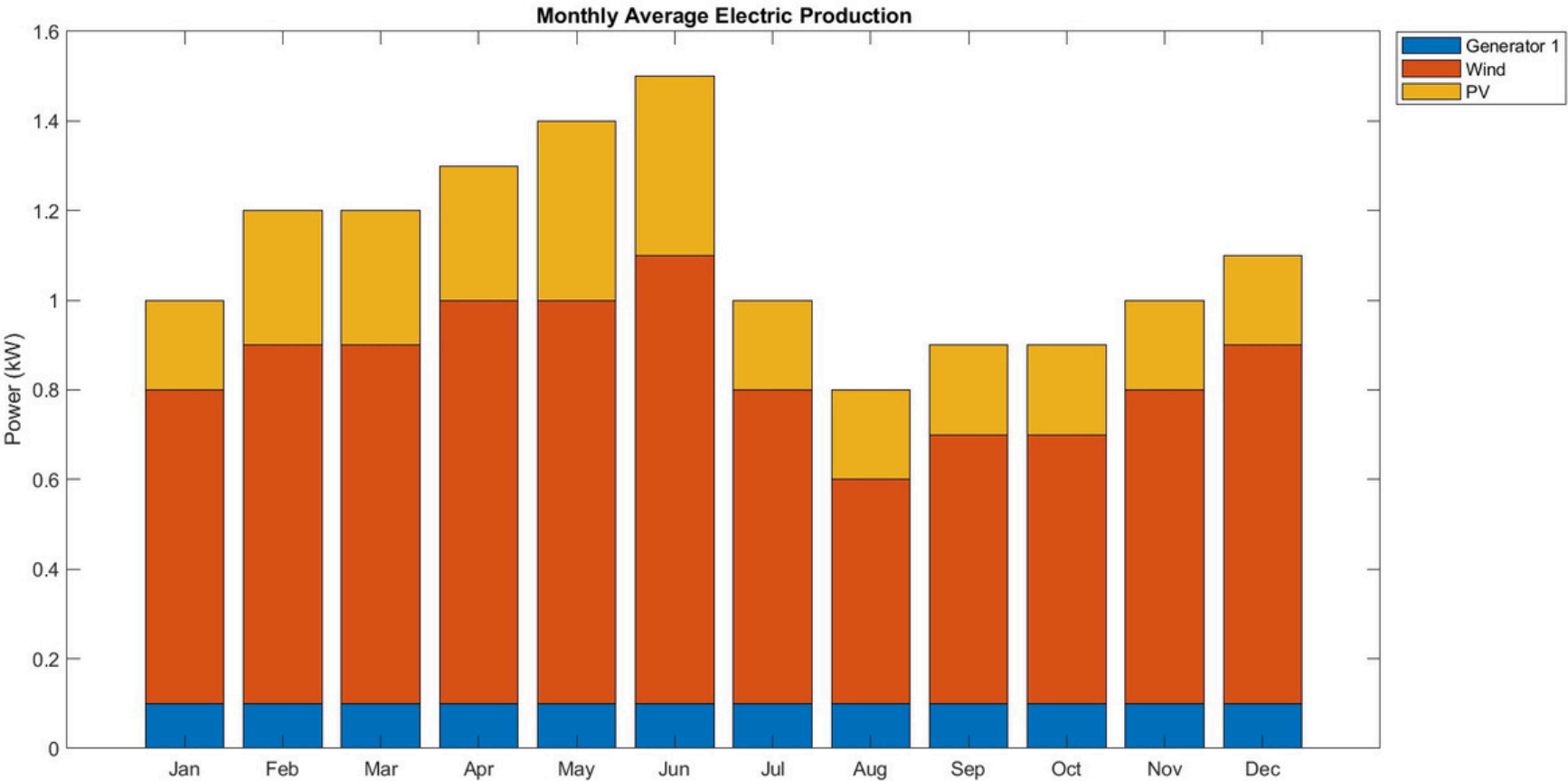


Fig 8.a - Annual power production by PV-Wind-Diesel-Battery

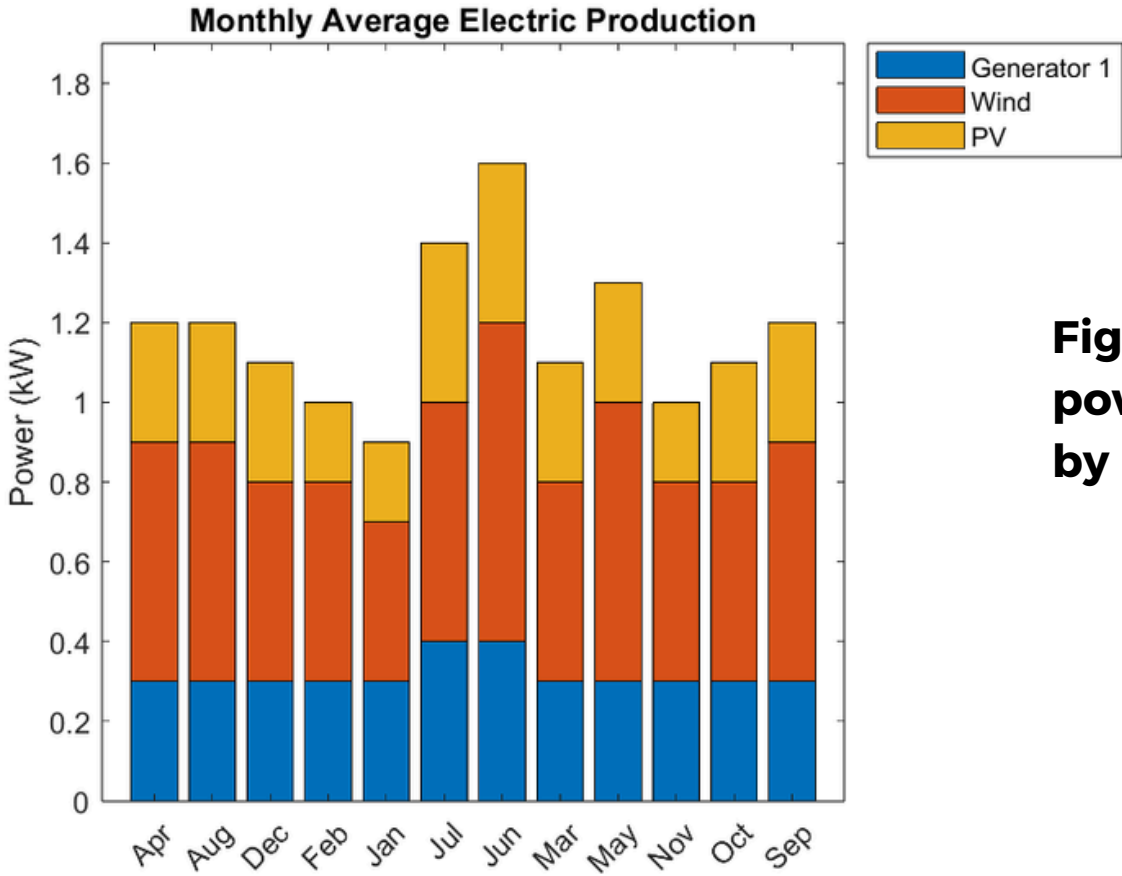


Fig 8.b- Annual power production by PV-Wind-DG

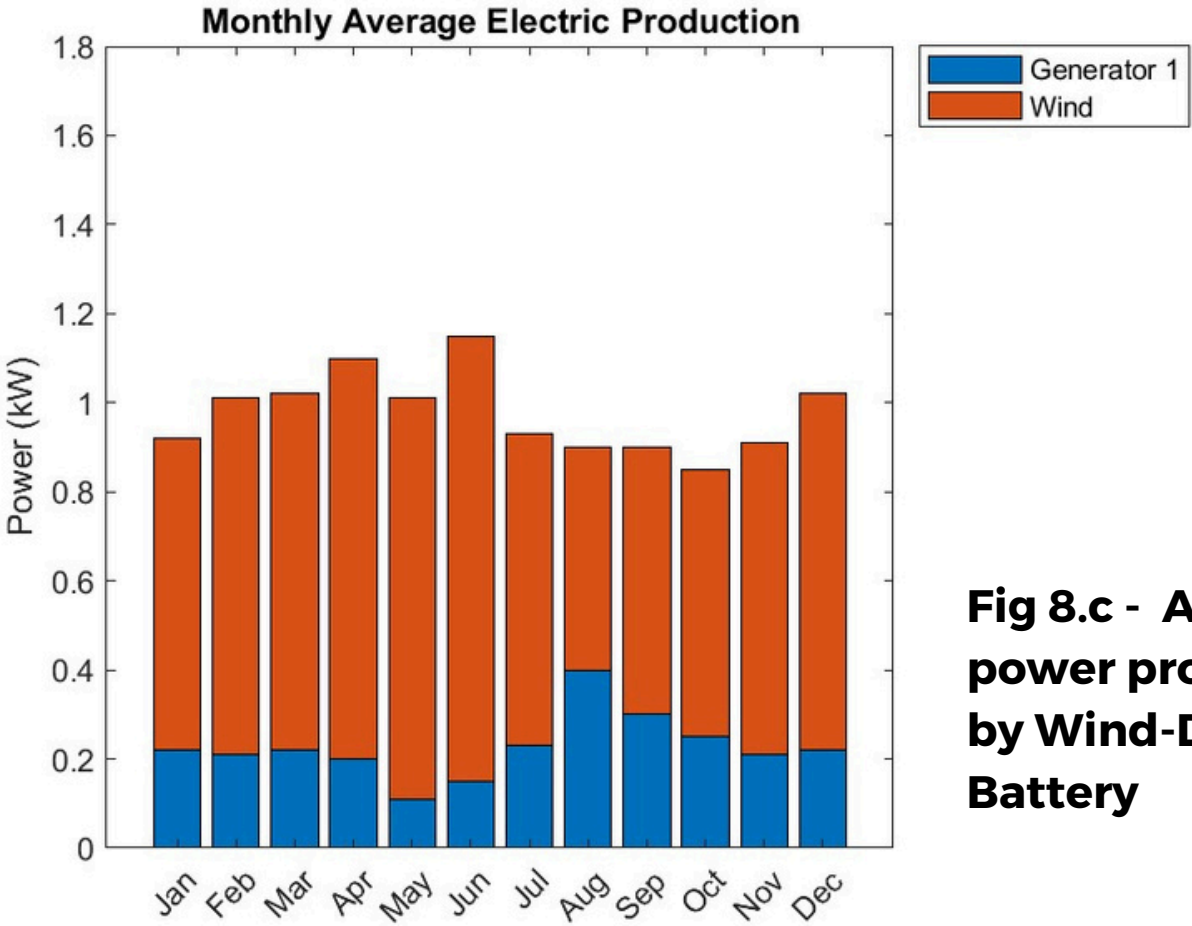


Fig 8.c - Annual power production by Wind-Diesel-Battery

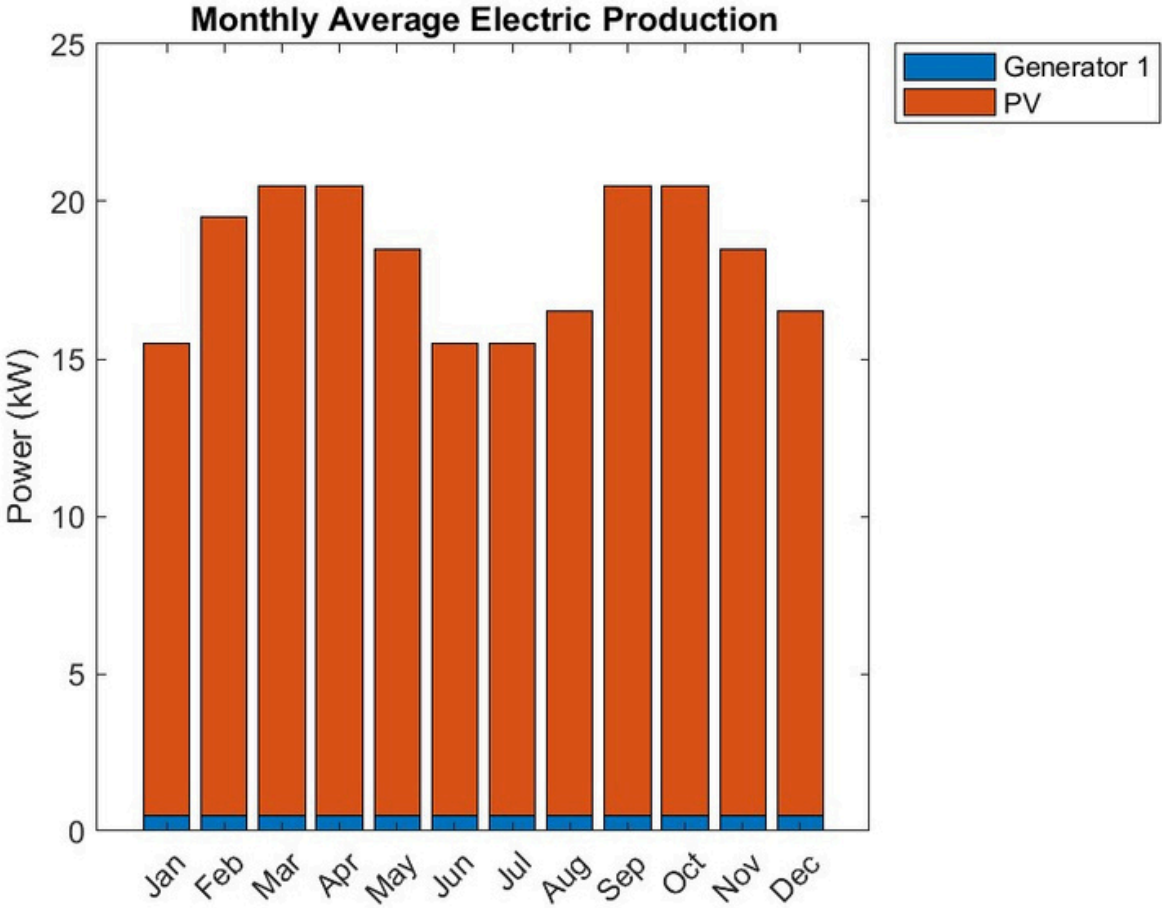


Fig 8.d- Annual power production by PV-Diesel

Problem Statement

Systems	COE (\$/kW h)	Total Power Generation (kW h/year)
PV -WIND-DIESEL-BATTERY	0.164	9799
WIND - DIESEL-BATTERY	0.215	9223
PV-DIESEL- BATTERY	0.266	5696
PV-WIND-DIESEL	0.293	8536
WIND-DIESEL	0.306	10,220
PV - DIESEL	0.709	166,469

FTable 5- Comparison of COE and power generation for different Hybrid System

- The main issues associated with all these configurations is the aim of no-carbon emission getting lost somewhere in the process
- To compensate the inadequacy of Solar and Wind systems , the researchers often go back to diesel generators. Hence, killing the purpose of usage of Renewable Energy Sources
- Also, in case of storage batteries being used the overall efficiency of the system is compromised big time
- So, the main problem is the lack of a Standalone system that is:
 - i) Off-Grid
 - ii) Capable Enough to Scale
 - iii) Net-zero Carbon Emission
- So **Optimal selection and design of off-Grid Hybrid renewable energy system** is needed to eradicate all these issues

The main objectives for this project are as follows:

- Firstly, appropriate site selection needs to be done so as to further plan the effective HRES that can fulfill the load demands.
- After the site selection, the domestic and commercial load trends are to be studied and modelled .
- The Model (Solar/PV + Wind Turbines + Hydrogen Fuel Cells) and the load requirements need to be formulated and techno-economic analysis is to be done using **HOMER** software.
- To Optimize the techno-economic outcomes of proposed HRES to get the least LCOE, LCOH, NPC and net operating cost and greater RF, ROI in HOMER.
- To perform a Sensitivity Analysis of the Optimized System account for variations in key parameters, including energy prices, subsidies, and financing mechanisms.

Methodology

- The load requirements of the village are to be formulated by the proposed model
- The excess electricity production from PV+WT system further needs to be used in electrolysis process for H₂
- The Techno-economic analysis of Polymer electrolyte membrane (PEM) fuel cells is then modelled with HOMER
- The optimized model is formulated for which the LCOE, LCOH, NPC are least
- Using HOMER sensitivity parameter such as Solar radiations and Wind flows are to be added to enhance the liability of the Hybrid renewable energy system.

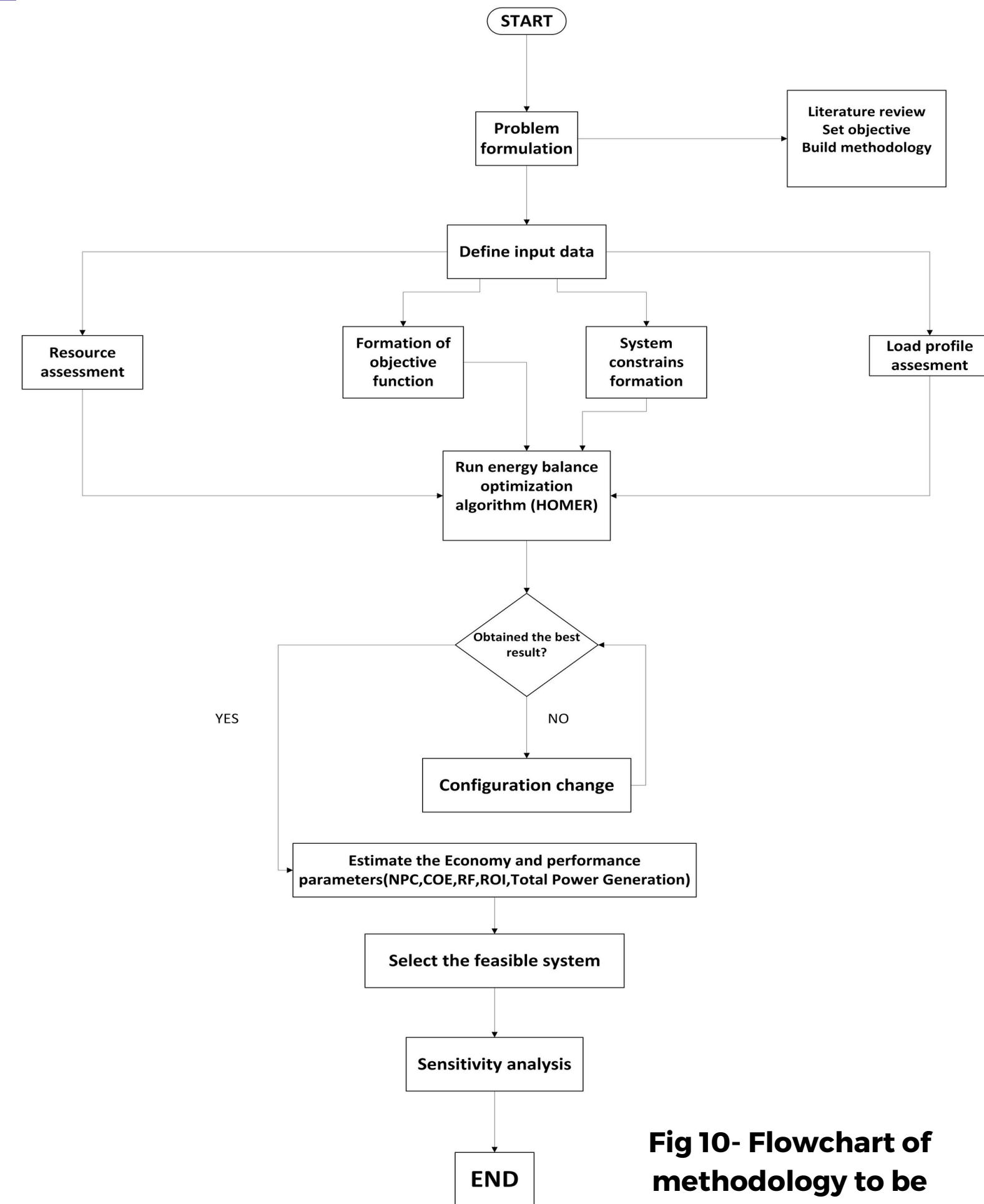
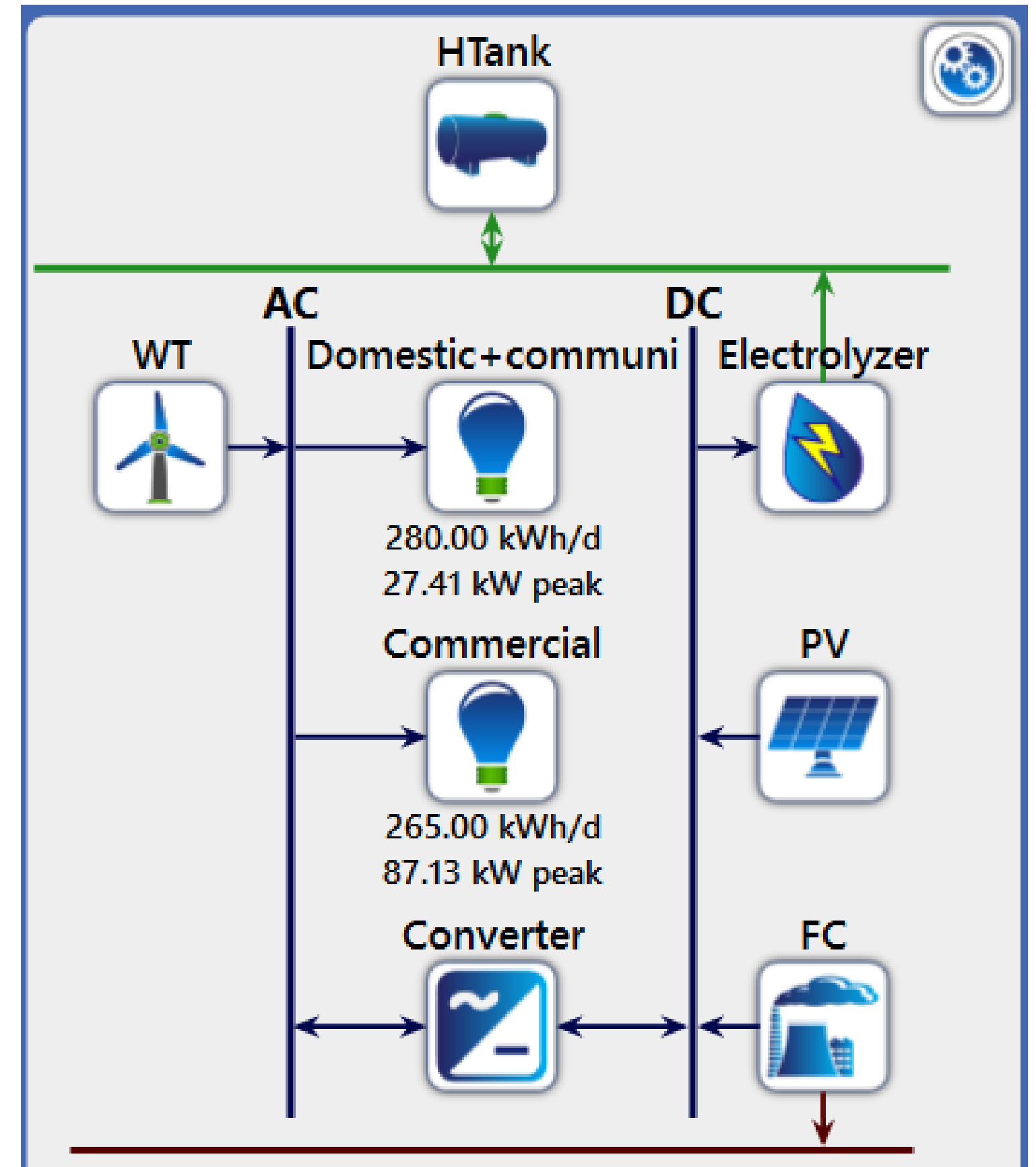
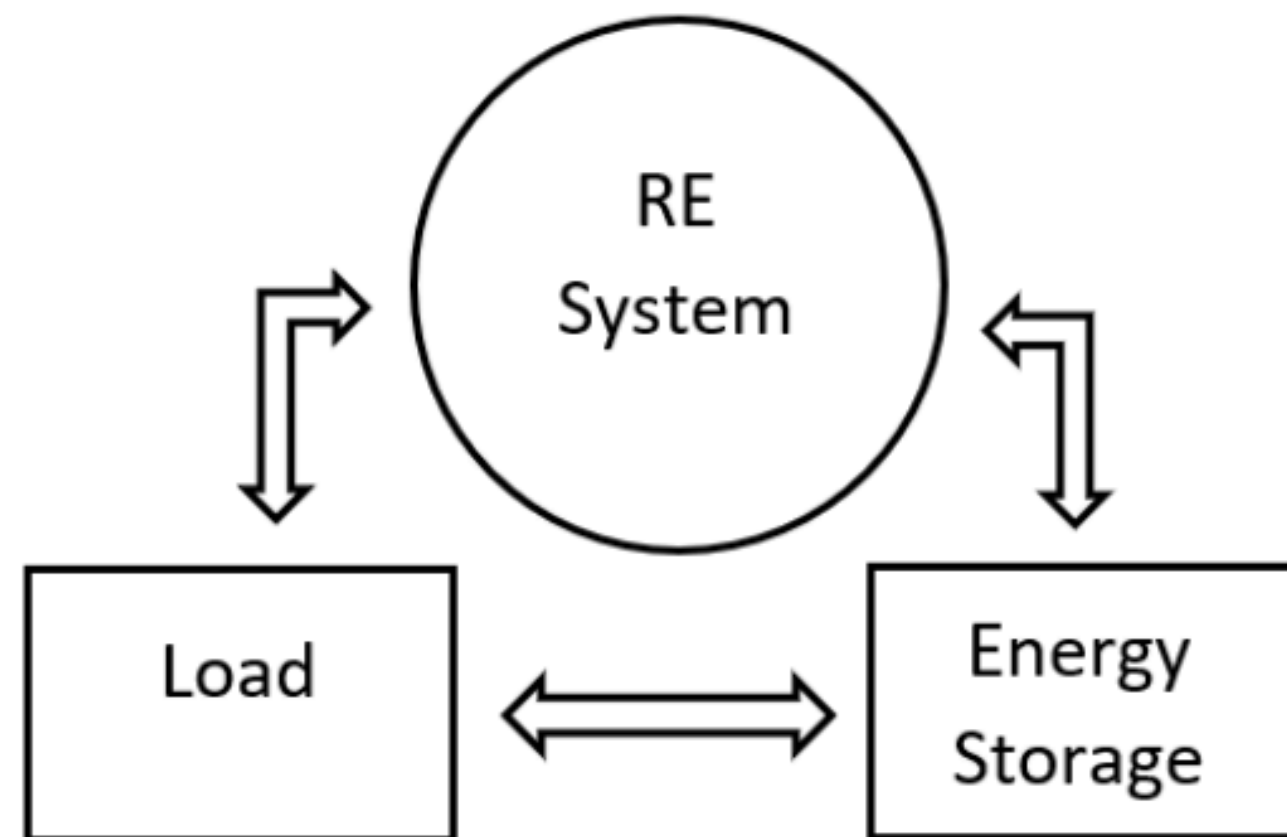


Fig 10- Flowchart of methodology to be applied

Proposed System

- HOMER(Hybrid Optimization of Multiple Energy Resources) Pro is firstly configured
- The Load demands are first met and the rest energy is supplied to Electrolyzer
- Converter is optimized as per HOMER default settings
- PV and Wind Turbines used are generic ones(1KW)
- Hydrogen tanks are also optimized at first then configured to bring the NPC value down



Limitations of HOMER Pro

- HOMER Pro doesn't have built in Fuel Cell for Stored Hydrogen
- It only has a Generic Electrolyzer
- The configurations of the Electrolyzer are limited
- HOMER Pro is a COE optimizer not sizing and electrical energy optimizer
- The Fuel Curve Slope(kg/h/kw) and the Intercept are to be entered manually
- The HRES need to be manually given the Capital, Replacement and O&M costs

WIND TURBINE 

Name: Abbreviation: Remove Copy To Library

Properties

Name: **Generic 1 kW**
Abbreviation: **G1**
Rated Capacity (kW): **1**
Manufacturer: **Generic**

Costs

Quantity	Capital (\$)	Replacement (\$)	O&M (\$/year)
Click here to add new item			

Multiplier:

Quantity Optimization

☒ HOMER Optimizer™
☐ Search Space
☐ Advanced

ELECTROLYZER 

Name: Abbreviation: Remove Copy To Library

Properties

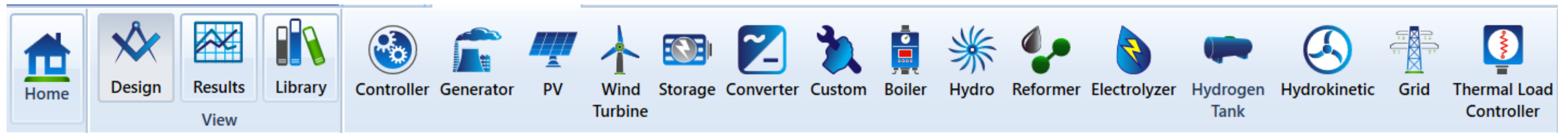
Name: **Generic Electrolyzer**
Abbreviation: **Electrolyzer**
Manufacturer: **Generic**
www.homerenergy.com
Notes:
This is a generic electrolyzer.

Costs

Capacity (kW)	Capital (\$)	Replacement (\$)	O&M (\$/year)
1	\$0.0	\$0.0	\$0.0
Click here to add new item			

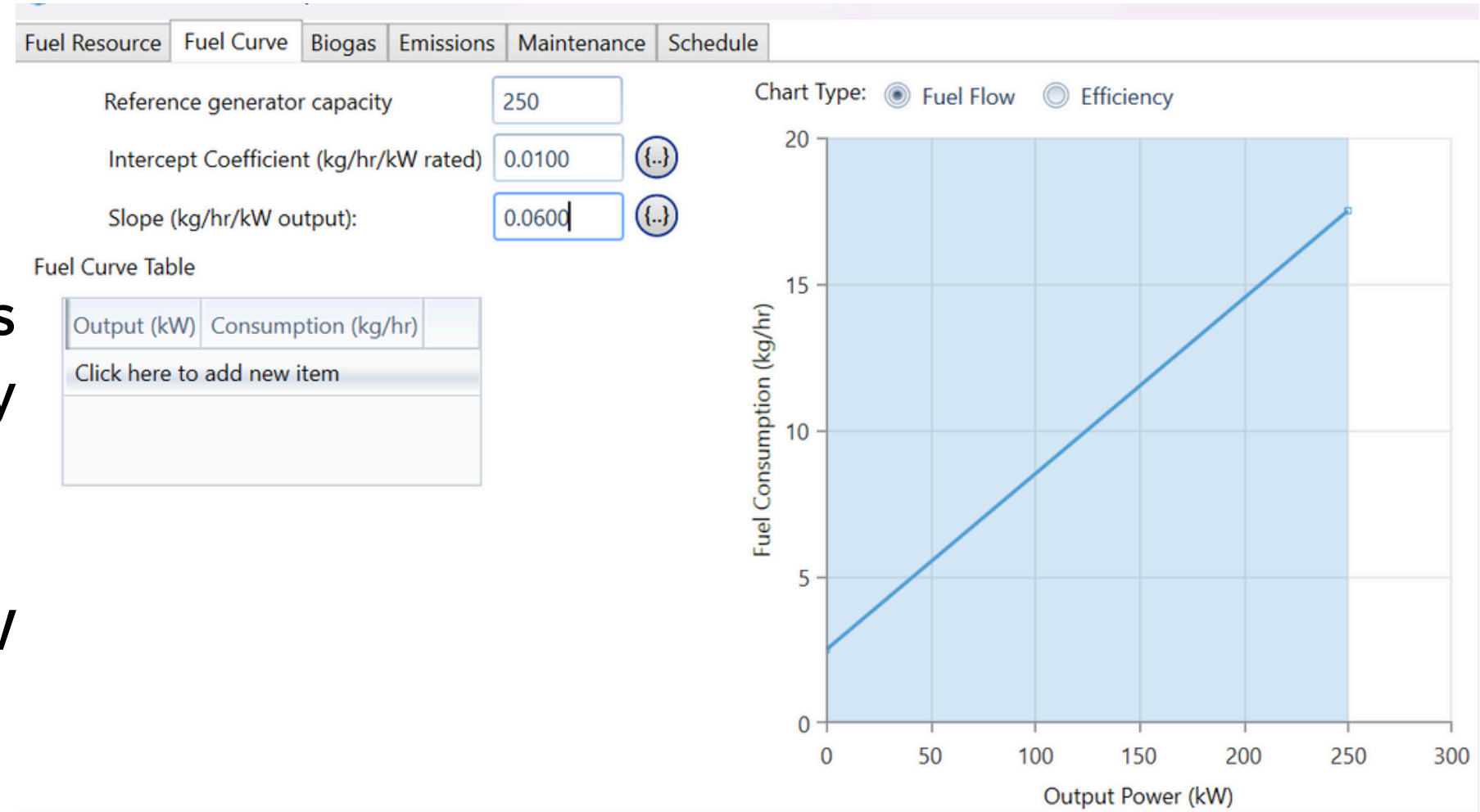
Multiplier:

Capacity Optimization



Configurations for HOMER Pro

- The Slope(kg/hr/kW) is set to 0.06 for stored hydrogen
- The Reference Fuel Cell capacity is 250KW
- The optimized results for PV and WT sizing is configured manually to reduce the excess electricity produced
- Electrolyzer rating is also manually limited to 80 KW to optimize the hydrogen production



RESULTS

Summary

Tables

Graphs

Calculation Report

Export...

Optimization Results

Left Double Click on a particular system to see its detailed Simulation Results.

☒ Categorized

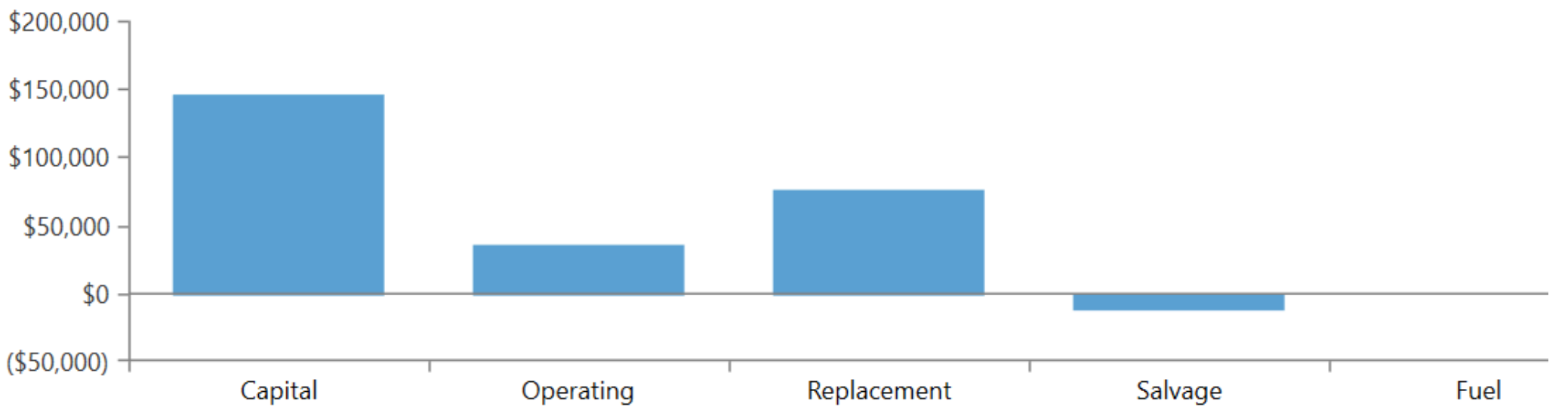
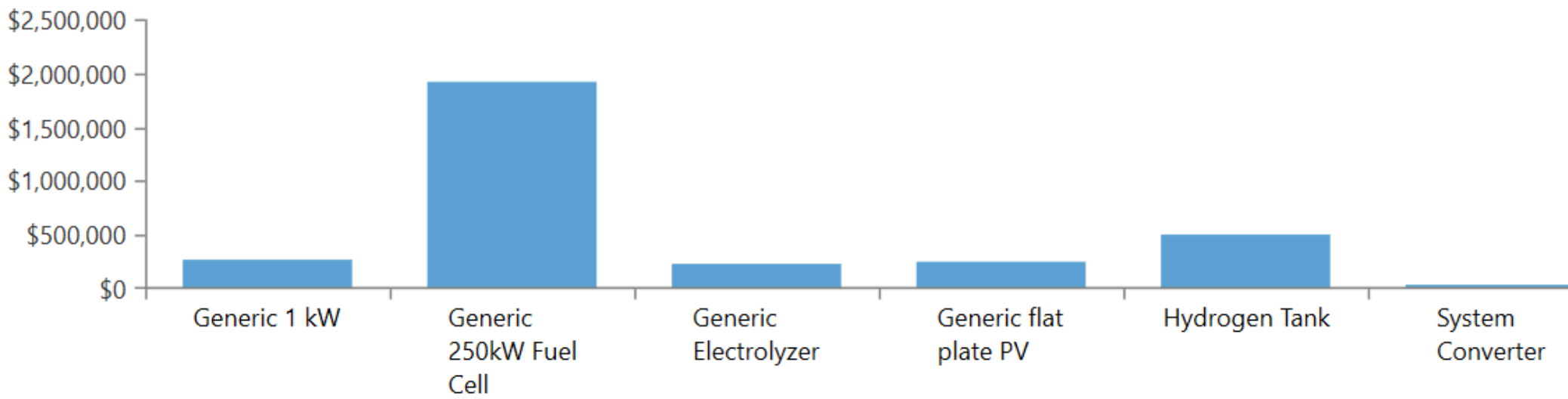
☐ Overall

Architecture												Cost				System				
						PV (kW)	G1	FC (kW)	Electrolyzer (kW)	HTank (kg)	Converter (kW)	Dispatch	NPC (\$)	COE (\$)	Operating cost (\$/yr)	Initial capital (\$)	Ren Frac (%)	Total Fuel (kg/yr)	Hours	Production (kWh)
						97.0	110	250	80.0	1,000	86.2	CC	\$3.22M	\$1.25	\$101,935	\$1.90M	100	3,607	6,661	60,112
						97.0		250	80.0	10,000	90.1	CC	\$7.93M	\$3.08	\$135,104	\$6.18M	100	6,000	8,726	100,000

Results

Economic-Financial Results

- The Fuel Cell(250kW) is the highest contributor of Capital Cost
- The system has a Total Expenditure of \$3,216,125.95(Capital+ Replacement+ O&M+ Fuel+ Salvage)
- Clearly the fuel cost is \$0 as the system is producing(Green Hydrogen) in the system itself
- The manual capacity optimization of PV and WT lead to the abrupt decline in the capital costs involved
- Economies of Scale analysis will further decrease the total system's cost



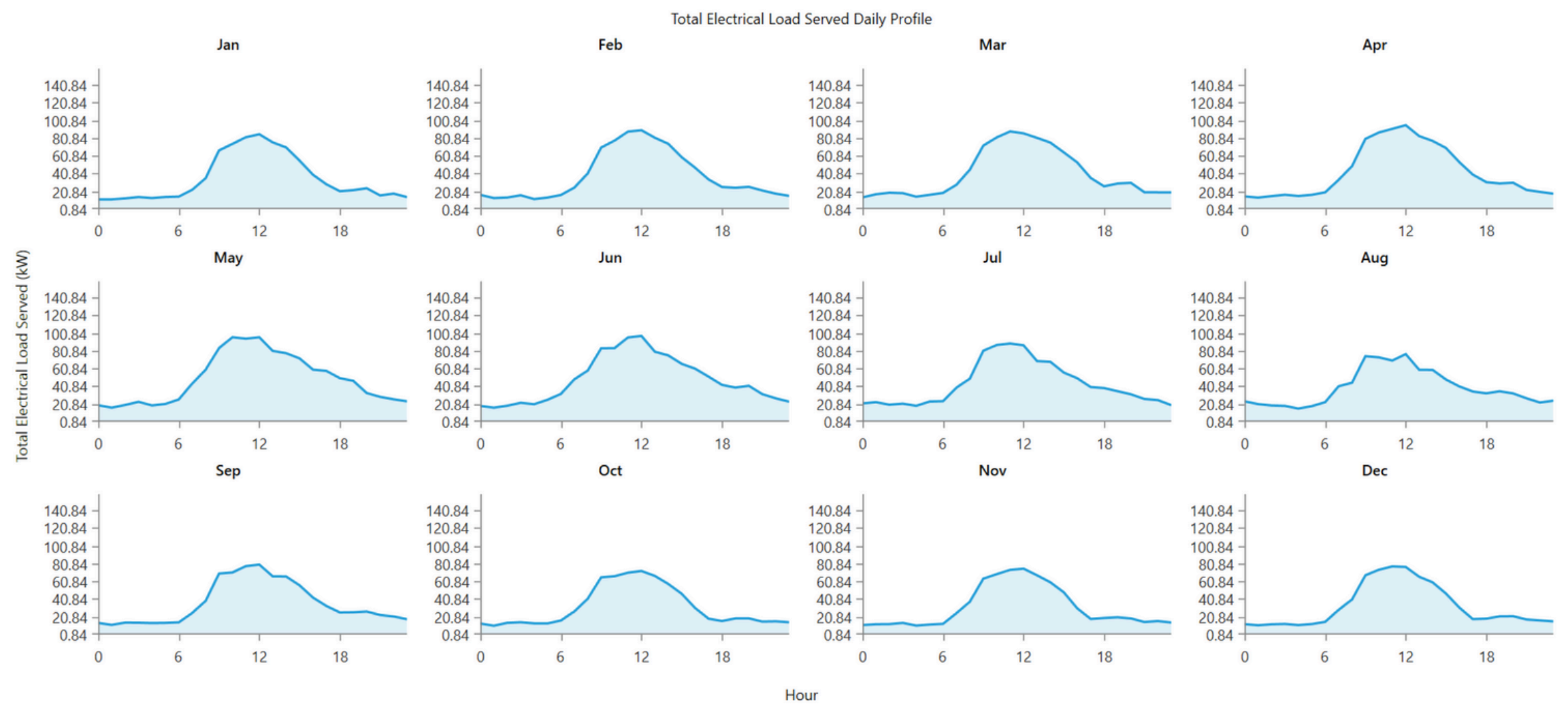
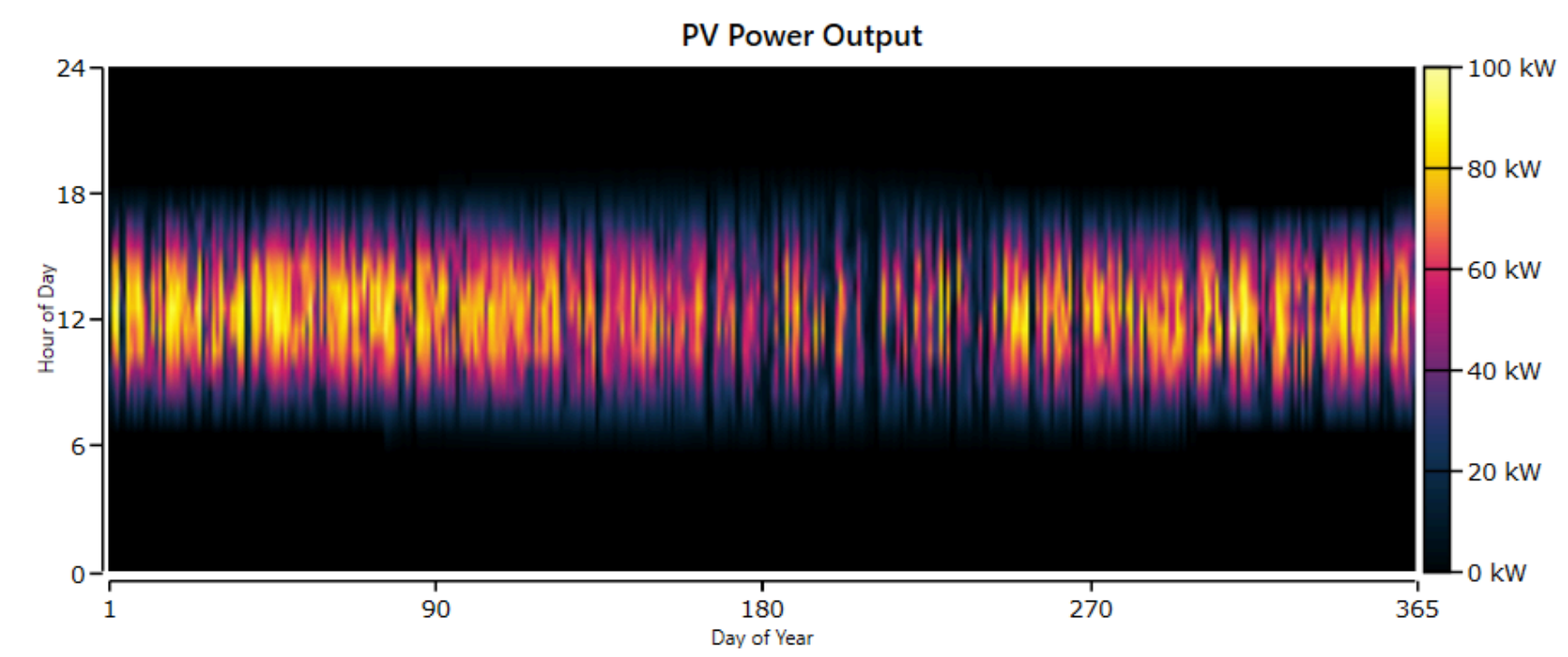
Component	Capital (\$)	Replacement (\$)	O&M (\$)	Fuel (\$)	Salvage (\$)	Total (\$)
Generic 1 kW	\$220,000.00	\$70,137.62	\$14,220.27	\$0.00	(\$39,527.05)	\$264,830.84
Generic 250kW Fuel Cell	\$750,000.00	\$844,465.09	\$430,550.94	\$0.00	(\$100,239.99)	\$1,924,776.04
Generic Electrolyzer	\$160,000.00	\$67,883.81	\$10,342.01	\$0.00	(\$12,776.42)	\$225,449.41
Generic flat plate PV	\$242,500.00	\$0.00	\$12,539.69	\$0.00	\$0.00	\$255,039.69
Hydrogen Tank	\$500,000.00	\$0.00	\$129.28	\$0.00	\$0.00	\$500,129.28
System Converter	\$25,854.62	\$10,969.44	\$11,141.20	\$0.00	(\$2,064.56)	\$45,900.70
System	\$1,898,354.62	\$993,455.95	\$478,923.39	\$0.00	(\$154,608.01)	\$3,216,125.95

Results

Solar PV Results

- Rated PV capacity for the proposed system is 97kW
- 43.1% of total load demands are met by PV system
- Generic flat plate PV is 1KW panel which has a \$2,55,040 total expenditure for this particular system

Quantity	Value	Units
Rated Capacity	97.0	kW
Mean Output	18.0	kW
Mean Output	431	kWh/d
Capacity Factor	18.5	%
Total Production	157,288	kWh/yr
Minimum Output	0	kW
Maximum Output	96.2	kW
PV Penetration	79.1	%
Hours of Operation	4,351	hrs/yr
Levelized Cost	0.125	\$/kWh

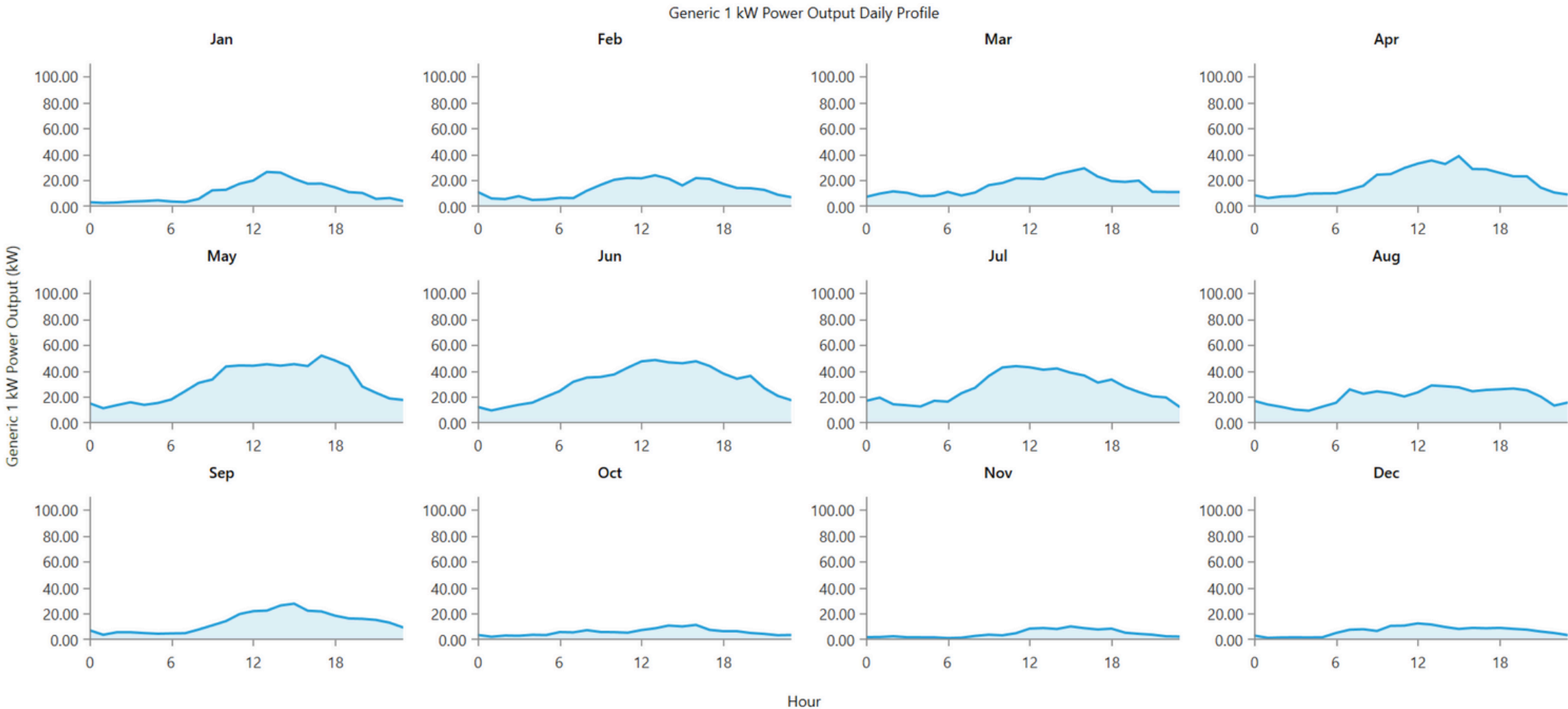
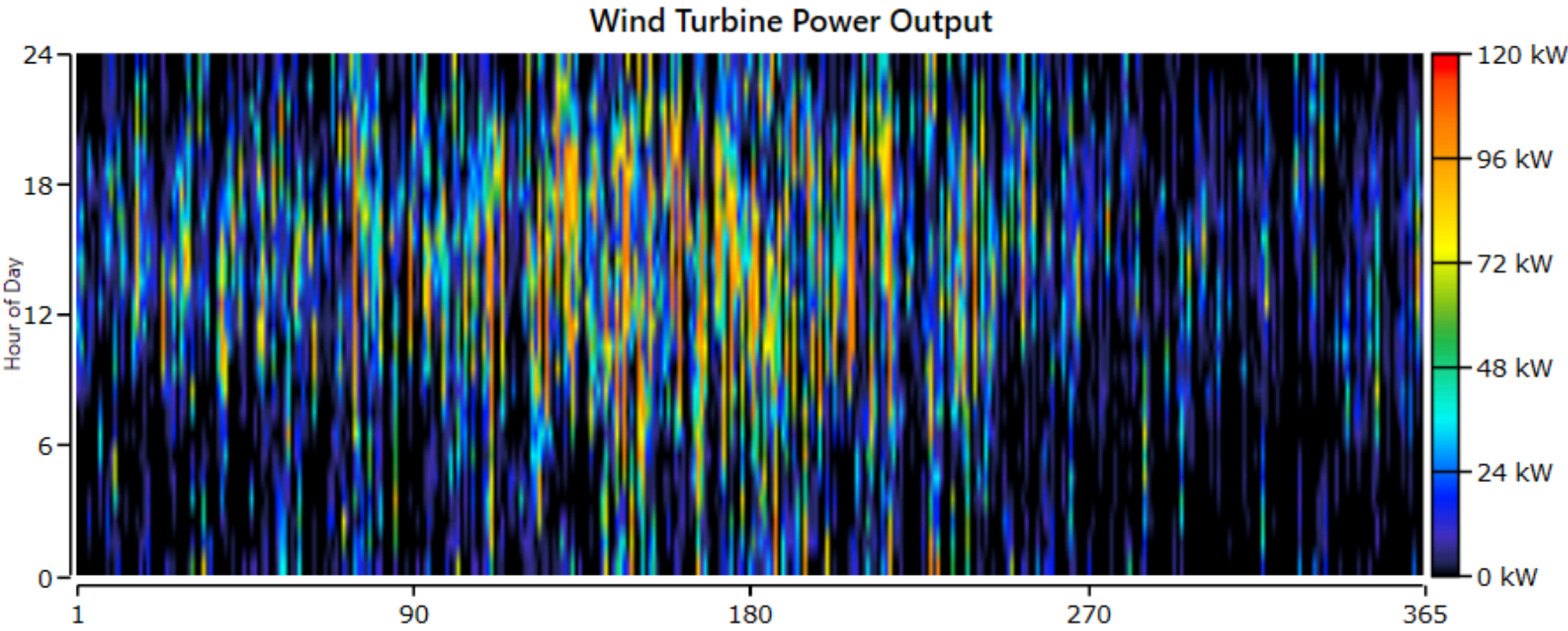


Results

Wind Turbine Results

- Total WT rated capacity for the proposed system is 110kW
- ~41% of total load demand is met by WT system
- Generic 1kW WT has a \$2,64,830 total expenditure for this particular system

Quantity	Value	Units
Total Rated Capacity	110	kW
Mean Output	16.9	kW
Capacity Factor	15.3	%
Total Production	147,611	kWh/yr
Minimum Output	0	kW
Maximum Output	110	kW
Wind Penetration	74.2	%
Hours of Operation	6,835	hrs/yr
Levelized Cost	0.139	\$/kWh

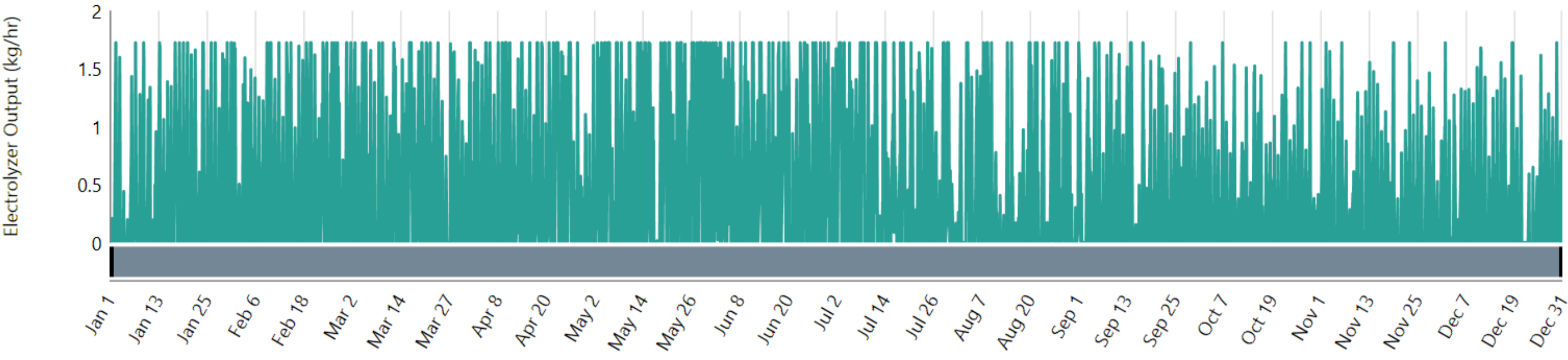
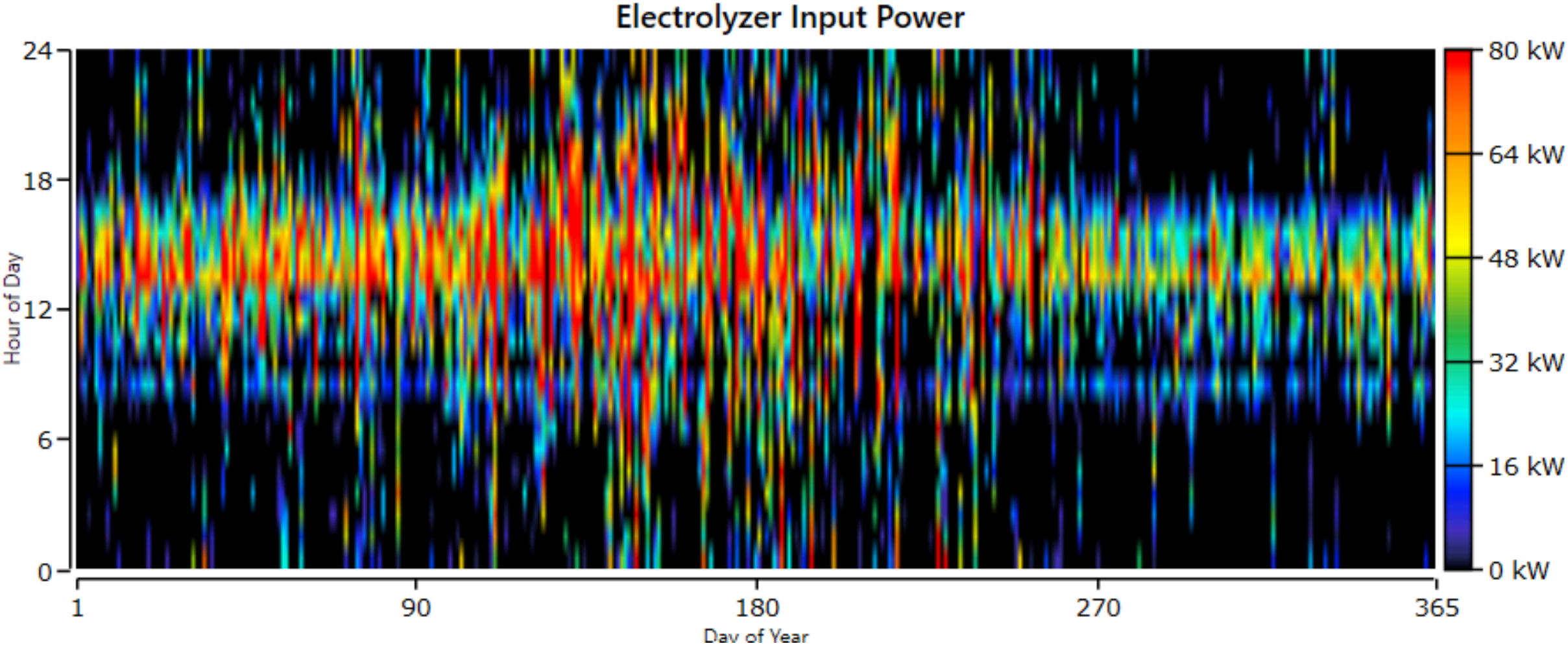


Results

Electrolyzer Results

- Manually limited the value of electrolyzer to 80kW
- Electrolyzer operates for 4202hr/yr
- The total load demand it adds to the system is 144,415 kWh/yr

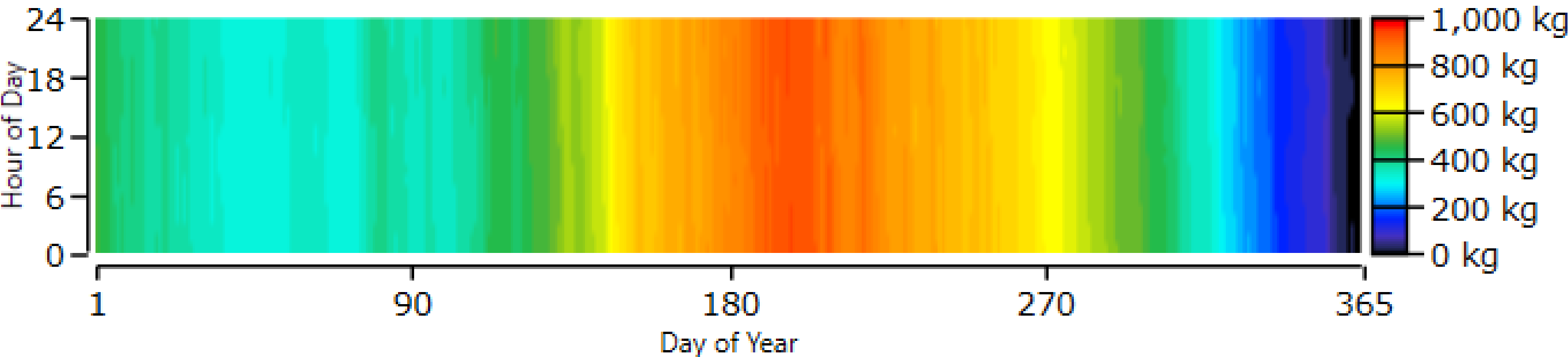
Quantity	Value	Units
Rated capacity	80.0	kW
Mean input	16.5	kW
Minimum input	0	kW
Maximum input	80.0	kW
Total input energy	144,415	kWh/yr
Capacity Factor	20.6	%
Hours of operation	4,202	hr/yr
Mean output	0.355	kg/hr
Minimum Output	0	kg/hr
Maximum Output	1.72	kg/hr
Total production	3,112	kg/yr
Specific consumption	46.4	kWh/kg



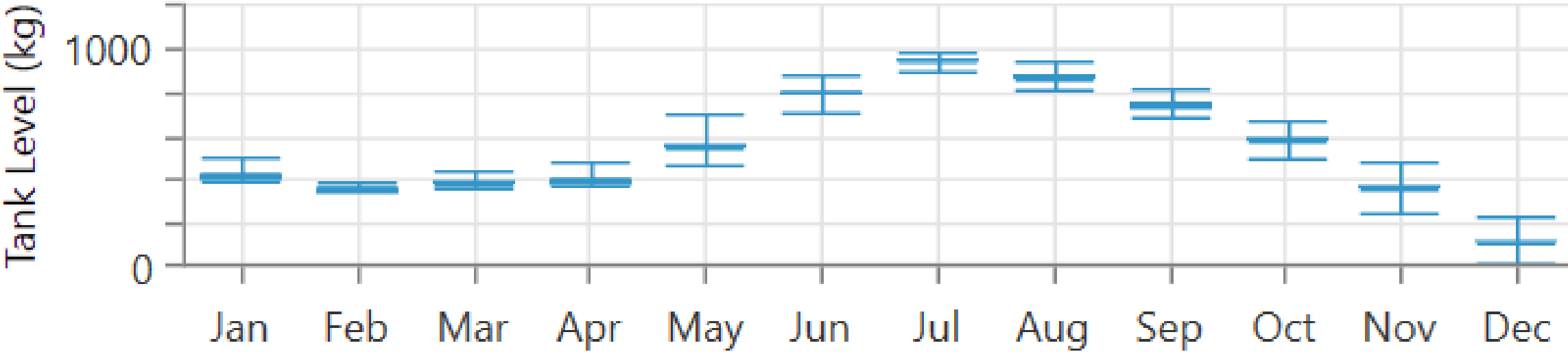
Results

Hydrogen Tank Results

- During the months of July-June the hydrogen utilization is less
- Seasonal conditions directly affect the stored hydrogen level
- More control of tank configurations will lead to better optimization



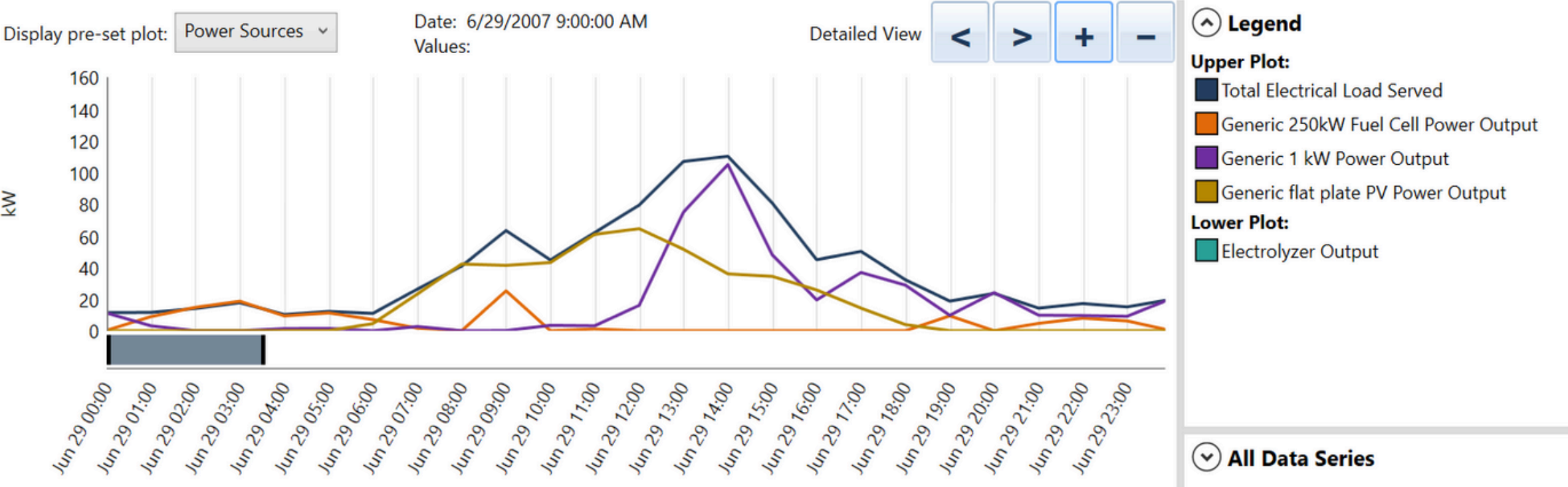
Quantity	Value	Units
Hydrogen storage capacity	1,000	kg
Energy storage capacity	33,333	kWh
Tank autonomy	1,468	hr
Content at beginning of year	500	kg
Content at end of year	5.31	kg



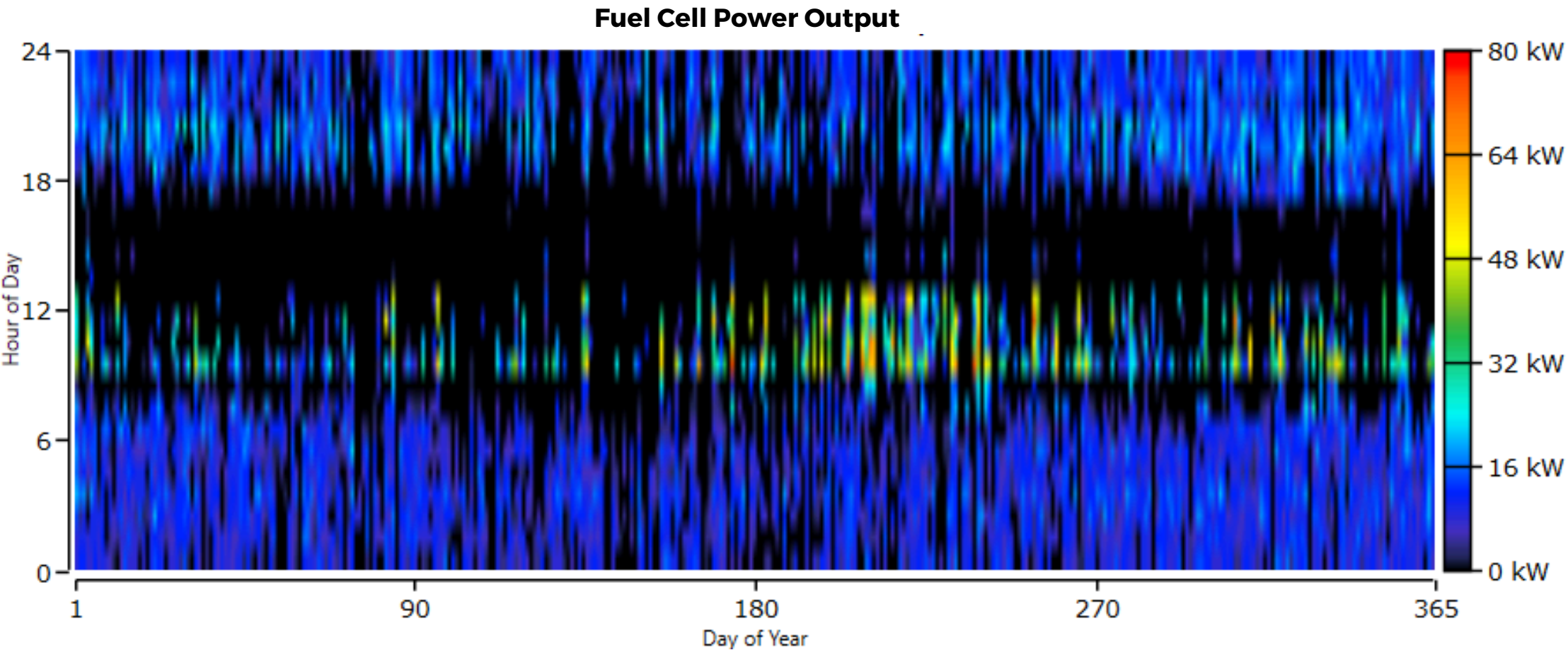
Results

Fuel Cell Results

- The intermittency of RES is clearly compensated by the hydrogen fed 250kW Fuel Cells
- At any hour of the day it can be observed that the fuel cell meets the load demand to keep LPSP 0



Quantity	Value	Units
Hours of Operation	6,661	hrs/yr
Number of Starts	629	starts/yr
Operational Life	7.51	yr
Capacity Factor	2.74	%
Fixed Generation Cost	17.5	\$/hr
Marginal Generation Cost	0	\$/kWh
Fuel Consumption	3,607	kg
Specific Fuel Consumption	0.0600	kg/kWh
Fuel Energy Input	120,224	kWh/yr
Mean Electrical Efficiency	50.0	%



Results

Solar-Wind with Battery Results

- PV (1kW): 100
- Generator G1 (3 kW): 26 unit
- 1kWh LA (Battery Units): 2,062 units
- Converter (1 kW): 109
- NPC (Net Present Cost): \$2.02M
- COE (Cost of Energy): \$0.786 per kWh

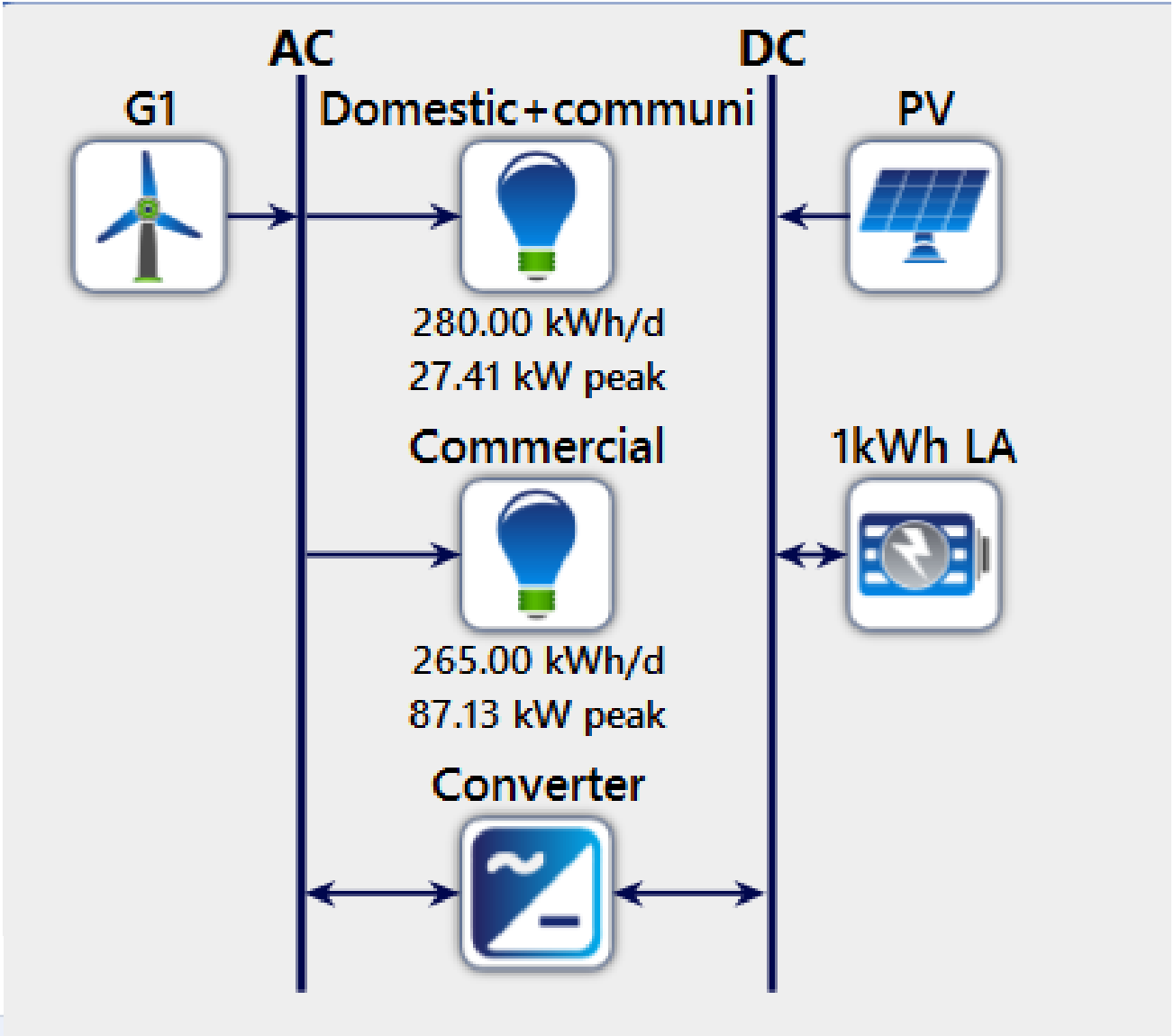
Export...

Optimization Results

Left Double Click on a particular system to see its detailed Simulation Results.

Architecture									Cost				System					
				PV (kW)		1kWh LA		Dispatch	NPC (\$)			Operating cost (\$/yr)		Initial capital (\$)		Ren Frac (%)		Total Fuel (L/yr)
				100	26	2,062	109	CC	\$2.02M	\$0.786		\$88,988		\$870,032	100		0	

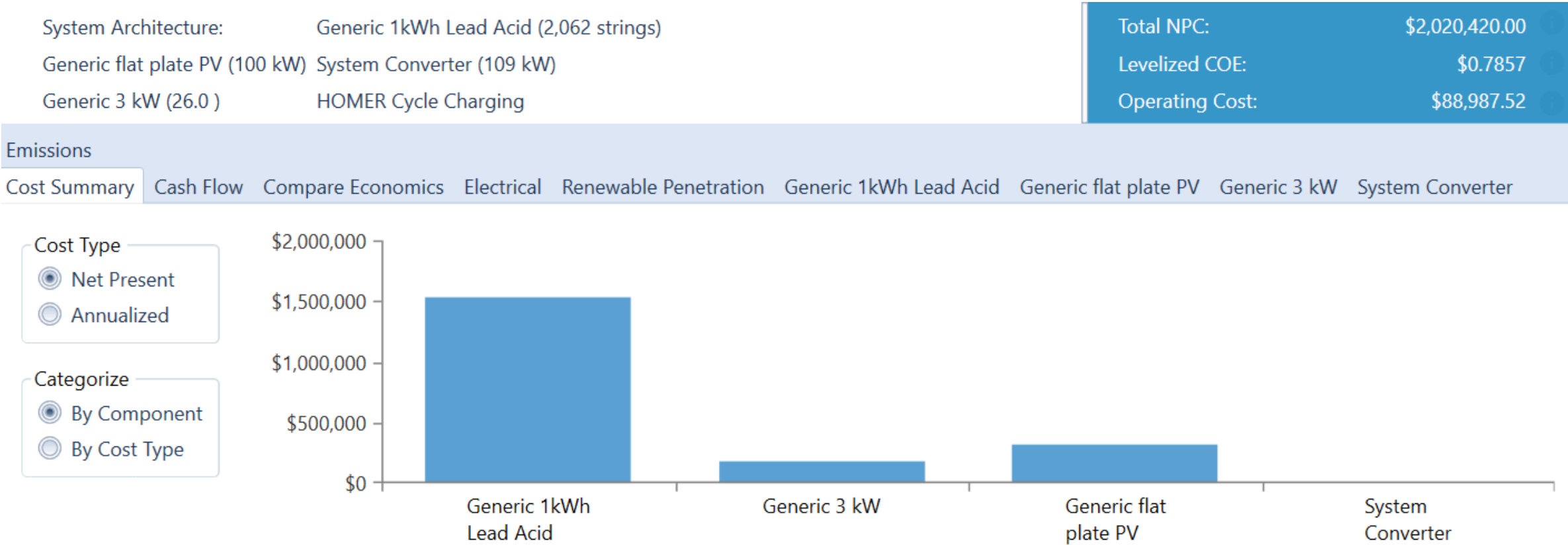
PV		G1		1kWh LA					Converter	
Capital Cost (\$)	Production (kWh/yr)	Capital Cost (\$)	Production (kWh/yr)	O&M Cost (\$)	Autonomy (hr)	Annual Throughput (kWh/yr)	Nominal Capacity (kWh)	Usable Nominal Capacity (kWh)	Rectifier Mean Output (kW)	Inverter Mean Output (kW)
300,000	162,153	156,000	149,899	1,560	54.5	62,217	2,064	1,238	2.04	13.4



- Initial Capital: \$870,032
- Operating Cost (per year): \$88,988
- Total Fuel Consumption: 0

Results

- The 1 kWh Lead Acid battery has the highest cost contribution among all components.
- It is followed by the system converter, flat plate PV system, and generic 3 kW units, in descending order of cost.



Consumption	kWh/yr	%
AC Primary Load	198,912	100
DC Primary Load	0	0
Deferrable Load	0	0
Total	198,912	100

The table represents the system's energy consumption data.

Results

- This balanced production highlights the system's reliance on both solar and wind energy sources for meeting the full demand.
- Solar contributing 52% and wind generating 48% of the total load.
- Conversion Losses: During the conversion process, 13,868 kWh is lost, resulting in a total throughput of 62,217 kWh per year
- The battery efficiency is approximately 80.17%.

Production	kWh/yr	%
Generic flat plate PV	162,153	52.0
Generic 3 kW	149,899	48.0
Total	312,052	100

Quantity	Value	Units
Average Energy Cost	0	\$/kWh
Energy In	69,141	kWh/yr
Energy Out	55,648	kWh/yr
Storage Depletion	375	kWh/yr
Losses	13,868	kWh/yr
Annual Throughput	62,217	kWh/yr

Results

Comparative Analysis

Proposed System(PV+WT+HFC)

- LCOE is **\$1.25** considering the constraints HOMER Pro has
- Highest contributor in Total NPC is 250kW rated Fuel Cell used
- Due to absence of specialized Fuel Cell(PEM,AEM,SOEC) Electrolyzer the units are costing the system separately
- Fuel Cell efficiency in HOMER Pro is 50%
- For a tank of 1000kg of Hydrogen the Power stored is 33,333kWh

Battery System(PV+WT+Bat)

- Battery System has an LCOE of **\$0.75**
- Highest Contributor of Total NPC is 2062kWh battery used
- The configurations of battery are in such a way that the LPSP is least. Hence, the excess energy generated in this case is high
- Battery used in the system have an efficiency of ~80%
- Energy Stored A total of 69,411 kWh per year is used for storage, produced by both solar and wind.

Results

Sensitivity Analysis

- The Cost of Electrolyzer is projected to reduce to half by coming 5 years, along with it's CAPEX, tank's cost will also be depreciated
- The multiplier function in Homer Pro is employed to do the Sensitivity analysis
- Significant improvement can be seen in the LCOE and NPC figures

RESULTS																						
Summary			Tables		Graphs		Calculation Report															
Export...		Export All...		Sensitivity Cases														Compare Economics		Column Choices...		
Left Click on a sensitivity case to see its Optimization Results.																						
Sensitivity			Architecture													Cost						
HTank Capital Cost Multiplier (*)	Electrolyzer Capital Cost Multiplier (*)	FC Lifetime (hours)								PV (kW)	WT	FC (kW)	Electrolyzer (kW)	HTank (kg)	Converter (kW)	Dispatch	NPC (\$)	COE (\$)	Operating cost (\$/yr)	Ir		
0.250	0.250	100,000								97.0	110	250	80.0	1,000	86.2	CC	\$2.19M	\$0.852	\$60,987	\$		
0.500	0.250	100,000								97.0	110	250	80.0	1,000	86.2	CC	\$2.32M	\$0.901	\$60,987	\$		
1.00	0.250	100,000								97.0	110	250	80.0	1,000	86.2	CC	\$2.57M	\$0.998	\$60,987	\$		
0.250	0.250	200,000								97.0	110	250	80.0	1,000	86.2	CC	\$1.95M	\$0.759	\$42,428	\$		
0.500	0.250	200,000								97.0	110	250	80.0	1,000	86.2	CC	\$2.08M	\$0.808	\$42,428	\$		
1.00	0.250	200,000								97.0	110	250	80.0	1,000	86.2	CC	\$2.33M	\$0.905	\$42,428	\$		
0.250	0.250	50,000								97.0	110	250	80.0	1,000	86.2	CC	\$2.72M	\$1.06	\$101,935	\$		
0.500	0.250	50,000								97.0	110	250	80.0	1,000	86.2	CC	\$2.85M	\$1.11	\$101,935	\$		
1.00	0.250	50,000								97.0	110	250	80.0	1,000	86.2	CC	\$3.10M	\$1.20	\$101,935	\$		
Optimization Results																						
Left Double Click on a particular system to see its detailed Simulation Results.																						
Categorized Over																						
Architecture										Cost				System			FC					
							PV (kW)	WT	FC (kW)	Electrolyzer (kW)	HTank (kg)	Converter (kW)	Dispatch	NPC (\$)	COE (\$)	Operating cost (\$/yr)	Initial capital (\$)	Ren Frac (%)	Total Fuel (kg/yr)	Hours	Producti (kWh)	
							97.0	110	250	80.0	1,000	86.2	CC	\$2.19M	\$0.852	\$60,987	\$1.40M	100	3,607	6,661	60,112	

Results

Sensitivity Analysis

- Best case scenario is is with multipliers 0.25,0.25 and 200,000 hours for hydrogen tank, Electrolyzer and fuel cell respectively
- This LCOE is even better than the battery system, which had \$0.786
- Whereas the proposed model got the LCOE

Component	Capital (\$)	Replacement (\$)	O&M (\$)	Fuel (\$)	Salvage (\$)	Total (\$)
Generic 1 kW	\$220,000.00	\$70,137.62	\$14,220.27	\$0.00	(\$39,527.05)	\$264,830.84
Generic 250kW Fuel Cell	\$750,000.00	\$0.00	\$430,550.94	\$0.00	(\$25,060.00)	\$1,155,490.94
Generic Electrolyzer	\$40,000.00	\$67,883.81	\$10,342.01	\$0.00	(\$12,776.42)	\$105,449.41
Generic flat plate PV	\$242,500.00	\$0.00	\$12,539.69	\$0.00	\$0.00	\$255,039.69
Hydrogen Tank	\$125,000.00	\$0.00	\$129.28	\$0.00	\$0.00	\$125,129.28
System Converter	\$25,854.62	\$10,969.44	\$11,141.20	\$0.00	(\$2,064.56)	\$45,900.70
System	\$1,403,354.62	\$148,990.87	\$478,923.39	\$0.00	(\$79,428.02)	\$1,951,840.86

Capital Cost Multiplier (0.25 *)	Total NPC:	\$1,951,841.00
Lifetime (200,000.00 hours)	Levelized COE:	\$0.7592
Capital Cost Multiplier (0.25 *)	Operating Cost:	\$42,427.81

												
PV (kW)	WT	FC (kW)	Electrolyzer (kW)	HTank (kg)	Converter (kW)	Dispatch	NPC (\$)	COE (\$)	Operating cost (\$/yr)	Initial capital (\$)	Ren Frac (%)	Total Fuel (kg/yr)
97.0	110	250	80.0	1,000	86.2	CC	\$1.95M	\$0.759	\$42,428	\$1.40M	100	3,607

Conclusion and Future Work

- Khiriya Bharka village is powered with a HRES with LPSP of zero
- Nearly 3000kg of Green Hydrogen is produced and is then used to in Fuel Cells to compensate the intermittency of Solar and Wind system
- The generic Electrolyzer produces 3112kg/yr Green Hydrogen
- The Renewable Fraction is 100%
- Carbon Emissions are zero
- By economies of scale the levelized cost of electricity is further reduced from \$1.25 to \$0.759
- Scope and incentives in green hydrogen systems by Government and other bodies can be taken into consideration to further reduce the economic viabilities

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